6

Solving the Schrödinger equation in periodic solids

In the previous chapter we encountered density functional theory (DFT) which is extensively used for calculating the electronic structure of periodic solids. Aside from DFT, carefully designed potentials often allow accurate electronic structures to be obtained by simply solving the Schrödinger equation without going through the self-consistency machinery of DFT. In both approaches it is necessary to solve the Schrödinger equation and the present chapter focuses on this problem, although some comments on implementing a DFT self-consistency loop will be made.

The large number of electrons contained in a macroscopic crystal prohibits a direct solution of the Schrödinger equation for such a system. Fortunately, the solid has periodic symmetry in the bulk, and this can be exploited to reduce the size of the problem significantly, using *Bloch's theorem*, which enables us to replace the problem of solving the Schrödinger equation for an infinite periodic solid by that of solving the Schrödinger equation in a unit cell with a series of different boundary conditions – the so-called *Bloch boundary conditions*. Having done this, there remains the problem that close to the nuclei the potential diverges, whereas it is weak when we are not too close to any of the nuclei (interstitial region). We can take advantage of the fact that the potential is approximately spherically symmetric close to the nuclei, but further away the periodicity of the crystal becomes noticeable. These two different symmetries render the solution of the Schrödinger equation in periodic solids difficult. In this chapter we consider an example of an electronic structure method, the augmented plane wave (APW) method, which uses a spatial decomposition of the wave functions: close to the nuclei they are solutions to a spherical potential, and in the interstitial region they are plane waves satisfying the appropriate Bloch boundary conditions.

It is possible to avoid the problem of the deep potential altogether by replacing it by a weaker one, which leaves the interesting physical properties unchanged. This is done in the pseudopotential method which we shall also discuss in this chapter.

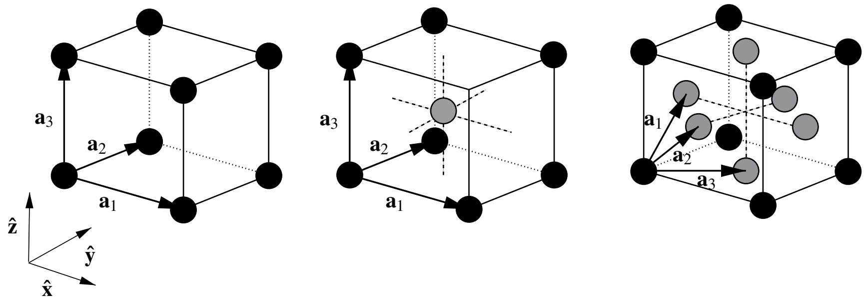


Figure 6.1. Lattice structure of the simple cubic (left), body-centred cubic (middle) and face-centred cubic (left) lattices with basis vectors.

Before going into these methods, we start with a brief review of the theory of electronic structure of solids. For further reading concerning the material in the first three sections of this chapter, we refer to general books on solid state physics [1, 2].

An excellent reference for computational band structures is the book by R. M. Martin [3].

6.1 Introduction: definitions

6.1.1 Crystal lattices

We consider crystals in which the atomic nuclei are perfectly ordered in a periodic lattice. Such a lattice, a so-called *Bravais* lattice, is defined by three basis vectors. The lattice sites **R** are given by the integer linear combinations of the basis vectors:

$$\mathbf{R} = \sum_{i=1}^{3} n_i \mathbf{a}_i, \quad n_i \text{ integer}. \tag{6.1}$$

Each cell may contain one or more nuclei – in the latter case we speak of a lattice with a basis. The periodicity implies that the arrangement of these nuclei must be the same within each cell of the lattice. Of course, in reality solids will only be approximately periodic: thermal vibrations and imperfections will destroy perfect periodicity and moreover, periodicity is destroyed at the crystal surface. Nevertheless, the infinite, perfectly periodic lattice is usually used for calculating electronic structure because periodicity facilitates calculations and a crystal usually contains large regions in which the structure is periodic to an excellent approximation.

Three common crystal structures, the simple cubic (sc), body-centred cubic (bcc) and face-centred cubic (fcc) structures, are shown in Figure 6.1.

6.1.2 Reciprocal lattice

A function which is periodic on a Bravais lattice can be expanded as a Fourier series with wave vectors $\mathbf{K}$ whose dot product with every lattice vector of the original lattice yields an integer times 2π:

$$\mathbf{R} \cdot \mathbf{K} = 2\pi n, \text{ integer } n. \tag{6.2}$$

The vectors $\mathbf{K}$ form another Bravais lattice – the *reciprocal lattice*. The basis vectors $\mathbf{b}_j$ of the reciprocal lattice are defined by

$$\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi \delta_{ij} \tag{6.3}$$

and an explicit expression for the $\mathbf{b}_j$ is

$$\mathbf{b}_j = 2\pi \varepsilon_{jkl} \frac{\mathbf{a}_k \times \mathbf{a}_l}{\mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3)}. \tag{6.4}$$

ε_{jkl} is the Lévi–Civita tensor, which is $+1$ for jkl an even permutation of $(1, 2, 3)$, and -1 for odd permutations.

In the reciprocal lattice, the first Brillouin zone is defined as the volume in reciprocal space consisting of the points that are closer to the origin than to any other reciprocal lattice point. A general wave vector $\mathbf{q}$ is usually decomposed into a vector $\mathbf{k}$ of the first Brillouin zone and a vector $\mathbf{K}$ of the reciprocal lattice:

$$\mathbf{q} = \mathbf{k} + \mathbf{K}. \tag{6.5}$$

For a finite *rectangular* lattice of size $L_x \times L_y \times L_z$, the allowed wave vectors $\mathbf{q}$ to be used for expanding functions defined on the lattice are restricted by the boundary conditions. A convenient choice is periodic boundary conditions in which functions are taken periodic within the volume of $L_x \times L_y \times L_z$. In that case, vectors in reciprocal space run over the following values:

$$\mathbf{q} = 2\pi \left(\frac{n_x}{L_x}, \frac{n_y}{L_y}, \frac{n_z}{L_z} \right) \tag{6.6}$$

with integer n_x, n_y and n_z.

6.2 Band structures and Bloch's theorem

We know that the energy spectra of electrons in atoms are discrete. If we place two identical atoms at a very large distance from each other, their atomic energy levels will remain unchanged. Electrons can occupy the atomic levels on either of both atoms and this results in a double degeneracy. On moving the atoms closer together, this degeneracy will be lifted and each level splits into two; the closer we move the atoms together, the stronger this splitting. Suppose we play the same game with three instead of two atoms: then the atomic levels split into three different ones, and so on. A solid consists of an infinite number of atoms moved close together and therefore each atomic level splits into an infinite number, forming a *band*. It is our aim to calculate these bands.

We shall now prove the famous Bloch theorem which says that the eigenstates of the Hamiltonian with a periodic potential are the same in the lattice cells located at $\mathbf{R}_i$ and $\mathbf{R}_j$ up to a phase factor $\exp[\mathrm{i}\mathbf{q}\cdot(\mathbf{R}_i - \mathbf{R}_j)]$ for some reciprocal vector $\mathbf{q}$. We shall see that a consequence of this theorem is that the energy spectra are indeed composed of bands.

We write the Schrödinger equation in reciprocal space. The potential V is periodic and it can therefore be expanded as a Fourier sum over reciprocal lattice vectors $\mathbf{K}$:

$$V(\mathbf{r}) = \sum_{\mathbf{K}} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}} V_{\mathbf{K}}. \tag{6.7}$$

An arbitrary wave function ψ can expanded as a Fourier series with wave vectors $\mathbf{q}$ allowed by the periodic boundary conditions (6.6):[1]

$$\psi(\mathbf{r}) = \sum_{\mathbf{q}} \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}} C_{\mathbf{q}}. \tag{6.8}$$

Writing $\mathbf{q} = \mathbf{k} + \mathbf{K}$, the Schrödinger equation reads (in atomic units)

$$\left[\frac{1}{2}(\mathbf{k}+\mathbf{K})^2 - \varepsilon\right] C_{\mathbf{k}+\mathbf{K}} + \sum_{\mathbf{K}'} V_{\mathbf{K}-\mathbf{K}'} C_{\mathbf{k}+\mathbf{K}'} = 0. \tag{6.9}$$

This equation holds for each vector $\mathbf{k}$ in the first Brillouin zone: in the equation, wave vectors $\mathbf{k}+\mathbf{K}$ and $\mathbf{k}+\mathbf{K}'$ are coupled by the term with the sum over $\mathbf{K}'$, but no coupling occurs between $\mathbf{k}+\mathbf{K}$ and $\mathbf{k}'+\mathbf{K}'$ for different $\mathbf{k}$ and $\mathbf{k}'$. Therefore, for each $\mathbf{k}$ we can, in principle, solve the eigenvalue equation (6.9) and obtain the energy eigenvalues ε and eigenvectors $\mathbf{C}_{\mathbf{k}}$ with components $C_{\mathbf{k}+\mathbf{K}}$, leading to wave functions of the form (see (6.8))

$$\psi_{\mathbf{k}}(\mathbf{r}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} \left(\sum_{K} C_{\mathbf{k}+\mathbf{K}} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}}\right). \tag{6.10}$$

The eigenvalues form a discrete spectrum for each $\mathbf{k}$. The levels vary with $\mathbf{k}$ and therefore give rise to energy bands. Equation (6.9) yields an infinite spectrum for each $\mathbf{k}$. We might attach a label n, running over the spectral levels, alongside the label $\mathbf{k}$, to the energy level ε: $\varepsilon = \varepsilon_{n\mathbf{k}}$.

We can rewrite (6.10) in a more transparent form. To this end we note that the expression in brackets in this equation is a periodic function in $\mathbf{r}$. Denoting this periodic function by $u_{\mathbf{k}}(\mathbf{r})$, we obtain

$$\psi_{\mathbf{k}}(\mathbf{r}) = \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{r}} u_{\mathbf{k}}(\mathbf{r}). \tag{6.11}$$

[1] For an infinite solid, the sum over $\mathbf{q}$ becomes an integral.

The eigenstates of the Hamiltonian can thus be written in the form of a plane wave times a periodic function. Equivalently, evaluating such a wave function at two positions, separated by a lattice vector $\mathbf{R}$, yields a difference of a phase factor $e^{i\mathbf{k}\cdot\mathbf{R}}$, according to our previous formulation of Bloch's theorem.

Electronic structure methods for periodic crystals are usually formulated in reciprocal space, by solving an equation like (6.9) in which the basis functions are plane waves labelled by reciprocal space vectors. In fact, Bloch's theorem allows us to solve for the full electronic structure in real space by considering only one cell of the lattice for each $\mathbf{k}$ and applying boundary conditions to the cell as dictated by the Bloch condition (6.11). In particular, each facet of the unit cell boundary has a 'partner' facet which is found by translating the facet over a lattice vector $\mathbf{R}$. The solutions to the Schrödinger equation should on both facets be equal up to factor $\exp(i\mathbf{k}\cdot\mathbf{R})$. These boundary conditions determine the solutions inside the cell completely. We see that we can try to solve the Schrödinger equation either in reciprocal space or in real space. For nonperiodic systems, real-space methods enjoy an increasing popularity [4–7].

6.3 Approximations

The Schrödinger equation for an electron in a crystal can be solved in two limiting cases: the nearly free electron approximation, in which the potential is considered to be weak everywhere, and the tight-binding approximation in which it is assumed that the states are tightly bound to the nuclei. Both methods aim to reduce the difficulty of the band structure problem and to increase the understanding of band structures by relating them to those of two different systems which we can easily describe and understand: free electrons and electrons in single-atom orbitals. The tight-binding method has led to many computational applications. We shall apply it to graphene and carbon nanotubes.

6.3.1 The nearly free electron approximation

It is possible to solve Eq. (6.9) if the potential is small, by using perturbative methods. This is called the nearly free electron (NFE) approximation. You might consider this to be inappropriate as the Coulomb potential is certainly not small near the nuclei. Surprisingly, the NFE bands closely resemble those of aluminium, for example! We shall see later on that the pseudopotential formalism provides an explanation for this.

The main results of the NFE are that the bands are perturbed by an amount which is quadratic in the size of the weak potential V except close to *Bragg planes*

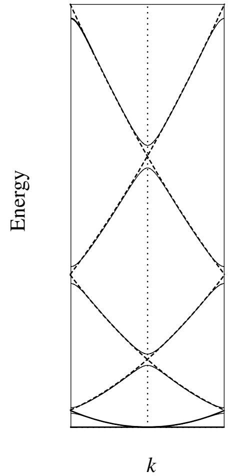


Figure 6.2. Nearly free electron spectrum for a periodic potential in one dimension.

consisting of reciprocal points $\mathbf{q}$ which satisfy

$$|\mathbf{q}| = |\mathbf{K} - \mathbf{q}|, \tag{6.12}$$

where $\mathbf{K}$ is a reciprocal lattice vector. At a Bragg plane, a band gap of size $2|V_{\mathbf{q}}|$ opens up. Figure (6.2) gives the resulting bands for the one-dimensional case. Figure (6.3) shows how well the bands in aluminium resemble the free electron bands.

6.3.2 The tight-binding approximation

The tight-binding (TB) approximation will be discussed in more detail as it is an important way for performing electronic structure calculations with many atoms in the unit cell. It naturally comes about when considering states which are tightly bound to the nuclei. The method is essentially a linear combination of atomic orbitals (LCAO) type of approach, in which the atomic states are used as basis orbitals. Let us denote these states, which are assumed to be available from some atomic electronic structure calculation, by $u_p(\mathbf{r} - \mathbf{R})$, in which the index p labels the levels of an atom located at $\mathbf{R}$. From these states, we build Bloch basis functions as

$$\phi_{p,\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{R}} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} u_p(\mathbf{r} - \mathbf{R}), \tag{6.13}$$

and a general Bloch state is a linear combination of these:

$$\phi_{\mathbf{k}}(\mathbf{r}) = \sum_p C_p(\mathbf{k}) \phi_{p,\mathbf{k}}(\mathbf{r}). \tag{6.14}$$

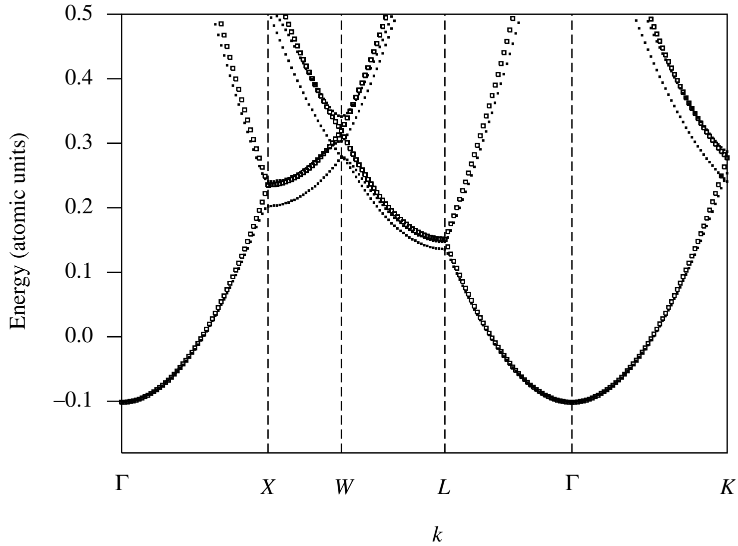


Figure 6.3. Band structure of aluminium. Also shown (open squares) is the free electron result. X, W etc. are special points in the Brillouin zone (see Section 6.5).

The coefficients $C_p(\mathbf{k})$ can be found from the variational principle. Applying the techniques of Chapter 3, we see that we must solve the generalised eigenvalue problem

$$\mathbf{HC}(\mathbf{k}) = E\mathbf{SC}(\mathbf{k}) \tag{6.15}$$

for the vector $\mathbf{C}(\mathbf{k})$ with components $C_p(\mathbf{k})$. The matrix elements of the Hamiltonian $\mathbf{H}$ and of the overlap matrix $\mathbf{S}$ (which depend on $\mathbf{k}$) are given by

$$H_{pq} = \langle \phi_{p,\mathbf{k}} | H | \phi_{q,\mathbf{k}} \rangle \text{ and} \tag{6.16a}$$

$$S_{pq} = \langle \phi_{p,\mathbf{k}} | \phi_{q,\mathbf{k}} \rangle. \tag{6.16b}$$

Writing out the matrix elements using (6.13) and the lattice periodicity, we obtain the following expressions:

$$H_{pq} = \sum_{\mathbf{R}} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \int \mathrm{d}^3 r\, u_p(\mathbf{r}-\mathbf{R}) H u_q(\mathbf{r}) \tag{6.17a}$$

$$S_{pq} = \sum_{\mathbf{R}} \mathrm{e}^{\mathrm{i}\mathbf{k}\cdot\mathbf{R}} \int \mathrm{d}^3 r\, u_p(\mathbf{r}-\mathbf{R}) S u_q(\mathbf{r}). \tag{6.17b}$$

As the states $u_p(\mathbf{r})$ are rather strongly localised near the nuclei, they will have virtually no overlap when centred on atoms lying far apart. This restricts the sums in (6.17) to only the first few shells of neighbouring atoms, sometimes only nearest neighbours. The numerical solution of the generalised eigenvalue problem $\mathbf{HC} = E\mathbf{SC}$ is treated in Chapter 3.

Of course, we may relax the condition that we consider atomic orbitals as basis functions, and extend the method to allow for arbitrary, but still localised, basis functions. This even works for the valence orbitals in metals, although in that case relatively many neighbours have to be coupled, so that the approach pays off only for large unit cells. We may use the tight-binding approach for fixed interatomic distances – for a tight-binding method in which the atoms are allowed to move, thereby requiring varying distances, see Chapter 9.

The tight-binding method comes in two flavours. The first is the *semi-empirical* TB method, in which only a few valence orbital basis functions are used. Their couplings are restricted to nearest neighbour atoms and the value of the couplings are fitted to either experimental data such as band gaps and band widths, or to similar data obtained using more sophisticated band structure calculations. Once satisfactory values have been obtained for the TB couplings, more complicated structures may be considered which are beyond reach of self-consistent DFT calculations.

We may also be more ambitious and use more TB parameters which are fitted to DFT Hamiltonians. This is particulary useful when the TB Hamiltonians were obtained using localised basis functions, such as Gaussian or Slater orbitals (see Chapter 4 for a dicussion of these basis sets). For DFT calculations, Slater type orbitals are becoming increasingly popular, as the reason for choosing Gaussians in Hartree–Fock calculations, i.e. the fact that integrals can be evaluated analytically, ceases to be relevant in DFT with its highly nonlinear exchange correlation potential. Therefore, the Hamiltonian naturally has a tight-binding form, and this means that it is *sparse*, that is, a small minority of the elements of the Hamiltonian are nonzero. Such a Hamiltonian allows for iterative methods to be used.

In the next subsection we consider an appealing application of the tight-binding method: graphene and carbon nanotubes.

6.3.3 Tight-binding calculation for graphene and carbon nanotubes

In this subsection, we calculate the band structure of a carbon nanotube within the tight-binding approximation. It is a very instructive exercise which is strongly recommended as an introduction to band structure calculations. Here we follow the discussion in Refs. [3, 8].

We assume that only elements of the Hamiltonian and overlap matrix coupling two atomic orbitals are relevant, and that three-point terms do not occur. This is the so-called *Slater–Koster* approximation [9]. We shall first apply this to graphene, which is a sheet consisting of carbon atoms ordered within a hexagonal lattice. This is not a Bravais lattice, but it can be described as a triangular Bravais lattice with a

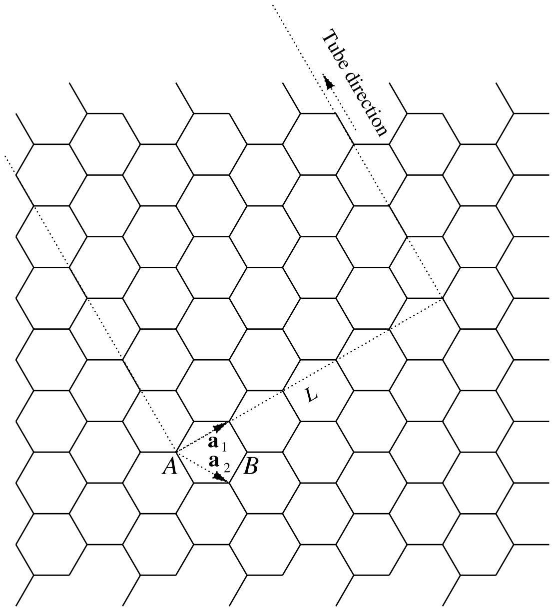


Figure 6.4. Hexagonal lattice of the graphene sheet with basis vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ indicated. The zigzag nanotube is also indicated.

basis consisting of two atoms, A and B (see Figure 6.4). The two basis vectors are

$$\mathbf{a}_1 = a\left(\frac{1}{2}\sqrt{3}, \frac{1}{2}\right); \quad \mathbf{a}_2 = a\left(\frac{1}{2}\sqrt{3}, -\frac{1}{2}\right). \tag{6.18}$$

where the lattice constant $a = 2.461$ Å. We only include nearest neighbour interactions.

The most relevant part of the band structure is the valence band – it turns out that this is formed by the π-orbitals which are built from the p_z-atomic orbitals. This leads us to taking only a single orbital per atom into account. Furthermore we keep only nearest neighbour matrix elements in the tight-binding matrices. We see from Figure 6.4 that an A-atom has nearest neighbours of the type B only.

The essential idea of using Bloch's theorem in calculating the band structure is to reduce the entire problem to that of the unit cell, which contains only two orbitals: the p_z orbitals of A and B. We must therefore calculate $H_{AA}(\mathbf{k}) = H_{BB}(\mathbf{k})$, $S_{AA}(\mathbf{k}) = S_{BB}(\mathbf{k})$, $H_{AB}(\mathbf{k}) = H^*_{BA}(\mathbf{k})$ and $S_{AB}(\mathbf{k}) = S^*_{BA}(\mathbf{k})$ *for each Bloch vector* $\mathbf{k}$. The spectrum is then given by the equation:

$$\begin{pmatrix} H_{AA}(\mathbf{k}) & H_{AB}(\mathbf{k}) \\ H_{BA}(\mathbf{k}) & H_{BB}(\mathbf{k}) \end{pmatrix} \begin{pmatrix} \psi_A(\mathbf{k}) \\ \psi_B(\mathbf{k}) \end{pmatrix} = E(\mathbf{k}) \begin{pmatrix} S_{AA}(\mathbf{k}) & S_{AB}(\mathbf{k}) \\ S_{BA}(\mathbf{k}) & S_{BB}(\mathbf{k}) \end{pmatrix} \begin{pmatrix} \psi_A(\mathbf{k}) \\ \psi_B(\mathbf{k}) \end{pmatrix}. \tag{6.19}$$

It follows that the energies $E(\mathbf{k})$ are given as

$$E_{\pm}(\mathbf{k}) = \frac{-(-2E_0 + E_1) \pm \sqrt{(-2E_0 + E_1)^2 - 4E_2E_3}}{2E_3}, \tag{6.20}$$

where

$$E_0 = H_{AA}S_{AA}; \qquad E_1 = S_{AB}H_{AB}^* + H_{AB}S_{AB}^*;$$
$$E_2 = H_{AA}^2 - H_{AB}H_{AB}^*; \qquad E_3 = S_{AA}^2 - S_{AB}S_{AB}^*.$$

The matrix element $H_{AA}(\mathbf{k})$ must be a real constant which does not depend on $\mathbf{k}$ and the same holds for $S_{AA}(\mathbf{k})$. We take the first to be 0 (with a suitable shift of the energy scale) and the second is 1 because of the normalisation of the orbitals.

You may verify that the off-diagonal elements have the form

$$H_{AB} = \gamma_0[\exp(i\mathbf{k} \cdot \mathbf{R}_1) + \exp(i\mathbf{k} \cdot \mathbf{R}_2) + \exp(i\mathbf{k} \cdot \mathbf{R}_3)] \tag{6.21}$$

with γ_0 a real constant independent of $\mathbf{k}$. The vectors $\mathbf{R}_1$ connect A to its three neighbours. For S_{AB} we find the same form but we call the constant s_0.

After some calculation, the energies are found as

$$E_{\pm}(\mathbf{k}) = \frac{\varepsilon_{2p} \mp \gamma_0\sqrt{f(\mathbf{k})}}{1 \mp s_0\sqrt{f(\mathbf{k})}}, \tag{6.22}$$

where

$$f(\mathbf{k}) = 3 + 2\cos(k_1) + 2\cos(k_2) + 2\cos(k_1 - k_2); \tag{6.23}$$

$$k_1 = \mathbf{k} \cdot \mathbf{a}_1, \quad k_2 = \mathbf{k} \cdot \mathbf{a}_2 \text{ etc.} \tag{6.24}$$

In the Brillouin zone, the point Γ is identified with $\mathbf{k} = (0, 0)$, and the point K is a vector of length $2\pi/a$ in the direction of $\mathbf{a}_2$; M is the point $2\pi/\sqrt{3}(1, 0)$. We can now plot the bands (i.e. the values $E_{\pm}(\mathbf{k})$ between M, Γ and K. The result is shown in Figure 6.5. Note that we find two energy values per atom. As we know that there should only be a single electron per p_z orbital, the Fermi energy must be the highest negative energy. We see that graphene is a metal: the bands touch each other precisely at the Fermi energy, so infinitesimal excitations are possible which yield nonzero momentum.

Now we turn to carbon nanotubes. These are graphene sheets rolled into a cylindrical form. There are many ways in which the two long ends can be glued together. These ways correspond to different strips which can be drawn on this sheet such that the two sides of the strip can be connected together smoothly. This is possible when the vector running perpendicularly across the sheet is an integer linear combination of the basis vectors $\mathbf{a}_1$ and $\mathbf{a}_2$. This is also indicated in Figure 6.4.

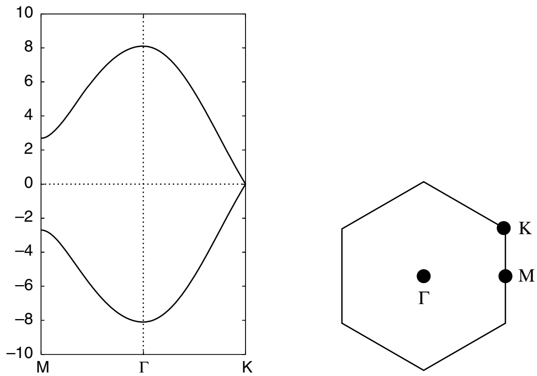


Figure 6.5. Tight-binding band structure of graphene. The points of the Brillouin zone are shown on the right hand side.

We speak now of an (n, m) nanotube where n and m are the integer coefficients. We restrict ourselves here to the case where $m = 0$. Such tubes are called 'zigzag' tubes because a circle running around the tube consists of a zigzag structure of nearest neighbour bonds.

A carbon nanotube is a one-dimensional object – therefore the states are labelled by a Bloch vector *along* the tube. We neglect effects due to the curvature of the sheet, which alter the interactions. For large tubes (i.e. tubes with a large diameter) this is a good approximation. Across the tube, the wave function *must be periodic*. The difference from a periodic cell in a periodic crystal must be emphasised here. In a crystal, the potential and the density are periodic, but the wave function (in general) is not. In the present case the wave function must match onto itself across the tube – hence it is really periodic. This implies that the transverse component of the wave vector must be $2\pi j/L$ for a tube circumference L and integer j. For each n we find an energy value, that is, for each fixed longitudinal k-vector we find a discrete energy spectrum.

For this case, the period along the tube is $a\sqrt{3}$. This means that the longitudinal Brillouin zone runs up to $k = \pi/(a\sqrt{3})$. This point is denoted as X. The transverse period is given as $L = na$. In order to calculate the band structure, we perform a loop over the longitudinal k-vector. For each such vector we run over the possible transverse k-vectors (values $2\pi j/L$ where j lies between 0 and n). We calculate the two energies in (6.20), and plot these as a function of the longitudinal k. The result is shown in Figure 6.6 for a tube with an odd and an even number of orbitals. Note the difference between the two: one is a metal, the other an insulator. In reality, the even tube has a small gap due to the curvature with respect to the graphene case. Tubes of another type, the so-called arm-chair tubes, characterised by $m = n$, are always metallic. The reader is invited to investigate that case.

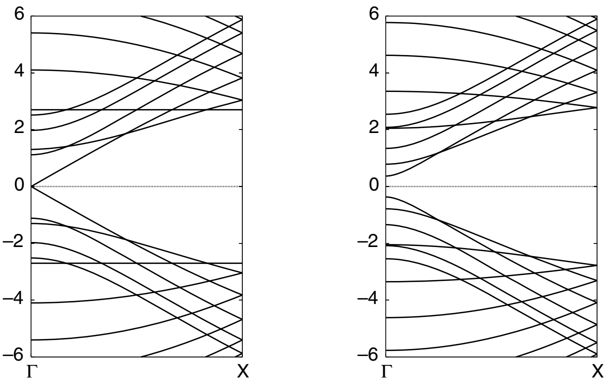


Figure 6.6. Energy bands for a (12, 0) and a (13, 0) tube. In reality, both have an energy gap. That of the even tube is very small. It is absent in the present analysis owing to the neglect of tube curvature.

6.4 Band structure methods and basis functions

Many band structure methods exist and they all have their own particular features. A distinction can be made between *ab initio* methods, which use no experimental input, and *semi-empirical* methods, which do. The latter should not be considered as mere fitting procedures: by fitting a few numbers to a few experimental data, many new results may be predicted, such as full band structures. Moreover, the power of *ab initio* methods should not be exaggerated: there are always approximations inherent to them, in particular the reliance on the Born–Oppenheimer approximation separating the electronic and nuclear motion, and often the local density approximation for exchange and correlation.

In *ab initio* methods, the potential is usually determined self-consistently with the electron density according to the DFT scheme. Some methods, however, solve the Schrödinger equation for a *given*, cleverly determined potential designed to give reliable results. The latter approach is particularly useful for nonperiodic systems where many atoms must be treated in the calculation.

In a general electronic structure calculation scheme we must give the basis functions a good deal of attention since we know that by cleverly choosing the basis states we can reduce their number, which has a huge impact on the computer time needed as the latter is dominated by the $\mathcal{O}(N^3)$ matrix diagonalisation.

Two remarks concerning the potential in a periodic solid are important in this respect. First, the potential grows very large near the nuclei whereas it is (relatively) small in the interstitial region. There is no sharp boundary between the two regions, but it is related to the distance from the nucleus where the atomic wave function

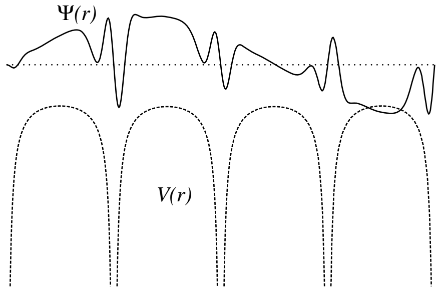


Figure 6.7. Valence state and Coulomb potential in a crystal.

becomes small. Second, the potential is approximately spherically symmetric near the nuclei, whereas at larger distances the crystal symmetry dominates.

Consider the valence state shown in Figure 6.7, together with the potential. Close to the nucleus, the valence state feels the strong spherically symmetric Coulomb potential and it will oscillate rapidly in this region. In between two nuclei, the potential is relatively weak and the orbital will oscillate slowly. The shape of this valence function can also be explained in a different way. Suppose that close to the nucleus, where the potential is essentially spherically symmetric, the valence wave function has s-symmetry (angular momentum quantum number $l = 0$), i.e. no angular dependence. There might also be core states with the same symmetry. As the states must be mutually orthogonal, the valence states must oscillate rapidly in order to be orthogonal to the lower states. A good basis set must be able to approximate such a shape using a limited number of basis functions.

For each Bloch wave vector, we need a basis set satisfying the appropriate Bloch condition. The most convenient Bloch basis set consists of plane waves:

$$\psi^{\mathrm{PW}}_{\mathbf{k}+\mathbf{K}}(\mathbf{r}) = \exp[\mathrm{i}(\mathbf{k}+\mathbf{K})\cdot\mathbf{r}]. \tag{6.25}$$

For a fixed Bloch vector **k** in the first Brillouin zone, each reciprocal lattice vector **K** defines a Bloch basis function for **k**. If we take a sufficient number of such basis functions into account, we can match any Bloch function for a Bloch vector **k**. However, Figure 6.7 suggests that we would need a huge number of plane waves to match our Bloch states because of the rapid oscillations near the nuclei. This is indeed the case: the classic example is aluminium for which it is estimated that 10^6 plane waves are necessary to describe the valence states properly [10]. Although plane waves allow efficient numerical techniques to be used, this number is very high compared with other basis sets, which yield satisfactory results with of the order of only 100 functions. Plane waves can only be used after cleverly transforming away the rapid oscillations near the nuclei, as in pseudopotential methods (Section 6.7).

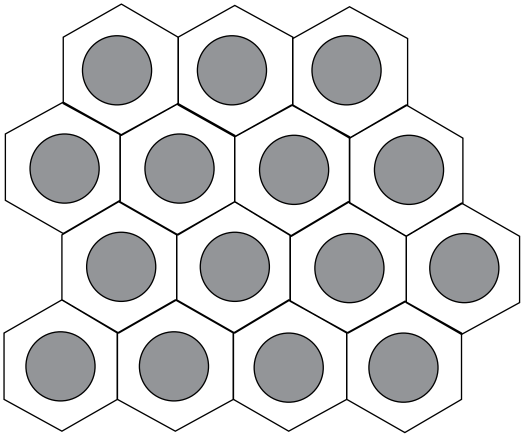


Figure 6.8. The muffin tin approximation.

In the next sections, we consider the augmented plane wave (APW) method and its linearised version, and construct a program for calculating the band structure of copper. In Section 6.7, the pseudopotential method will be considered with a few applications.

6.5 Augmented plane wave methods

6.5.1 Plane waves and augmentation

As we have just indicated, the problem in constructing basis sets starting from plane waves lies in the core region. Close to the nucleus, the potential is approximately spherically symmetric and this symmetry should be exploited in constructing the basis functions. In the augmented plane wave (APW) [11, 12] method, the nuclei are surrounded by spheres in which the potential is considered to be spherically symmetric and outside which the potential is constant, and usually taken to be zero. From a two-dimensional picture (Figure 6.8) it is clear why this approximation is known as the 'muffin tin' approximation.

Outside the muffin tin spheres, the basis functions are simple plane waves $e^{i\mathbf{q}\cdot\mathbf{r}}$. Inside the spheres, they are linear combinations of the solutions to the Schrödinger equation, evaluated at a predefined energy E. These linear combinations can be written as

$$\sum_{l=0}^{\infty}\sum_{m=-l}^{l} A_{lm}\mathcal{R}_l(r)Y_m^l(\theta,\varphi) \tag{6.26}$$

where the functions $\mathcal{R}_l(r)$ are the solutions of the radial Schrödinger equation with energy E:

$$-\frac{1}{2r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left[r^2\frac{\mathrm{d}\mathcal{R}_l(r)}{\mathrm{d}r}\right]+\left[\frac{l(l+1)}{2r^2}+V(r)\right]\mathcal{R}_l(r)=E\mathcal{R}_l(r) \tag{6.27}$$

which can be solved with high accuracy, and the $Y_m^l(\theta,\varphi)$ are the spherical harmonics. The expansion coefficients A_{lm} are found by matching the solution inside the muffin tin to the plane wave outside.

The Bloch wave $\exp(\mathrm{i}\mathbf{q}\cdot\mathbf{x})$ in the interstitial region is an exact solution to the Schrödinger equation at energy $q^2/2$ (in atomic units). The muffin tin solutions are numerically exact solutions of the Schrödinger equation at the energy E for which the radial Schrödinger equation has been solved. However, if we take this energy equal to $q^2/2$, the two solutions do not match perfectly. The reason is that the *general* solution in the interstitial region for some energy $E = q^2/2$ includes *all* wave vectors $\mathbf{q}$ with the same length. In the Bloch solution we take only one of these. If we want to solve the Schrödinger equation inside the muffin tins with the boundary condition imposed by this single plane wave solution then we would have to include solutions that diverge at the nucleus, which is physically not allowed.

An APW basis function contains a muffin tin solution with a definite energy E, and a Bloch wave $\exp(\mathrm{i}\mathbf{q}\cdot\mathbf{x})$ in the interstitial region. It turns out to be possible to match the amplitude of the wave function across the muffin tin sphere boundary. In order to carry out the matching procedure at the muffin tin boundary we expand the plane wave in spherical harmonics [13]:

$$\exp(\mathrm{i}\mathbf{q}\cdot\mathbf{r})=4\pi\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\mathrm{i}^l j_l(qr)Y_m^{l\,*}(\theta_{\mathbf{q}},\varphi_{\mathbf{q}})Y_m^l(\theta,\varphi) \tag{6.28}$$

where r, θ and φ are the polar coordinates of $\mathbf{r}$ and q, $\theta_{\mathbf{q}}$ and $\varphi_{\mathbf{q}}$ those of $\mathbf{q}$. To keep the problem tractable, we cut all expansions in lm off at a finite value for l:

$$\sum_{l=0}^{\infty}\sum_{m=-l}^{l}\rightarrow\sum_{l=0}^{l_{\max}}\sum_{m=-l}^{l} \tag{6.29}$$

From now on, we shall denote these sums by $\sum_{lm}$.

The matching condition implies that the coefficients of the Y_m^l must be equal for both parts of the basis function, (6.26) and (6.28), as the Y_m^l form an orthogonal set over the spherical coordinates. This condition fixes the coefficients A_{lm}, and we arrive at

$$\psi_{\mathbf{q}}^{\mathrm{APW}}(\mathbf{r})=4\pi\sum_{lm}\mathrm{i}^l\left[\frac{j_l(qR)}{\mathcal{R}_l(R)}\right]\mathcal{R}_l(r)Y_m^{l\,*}(\theta_{\mathbf{q}},\varphi_{\mathbf{q}})Y_m^l(\theta,\varphi) \tag{6.30}$$

for the APW basis function inside the sphere.

Summarising the results so far, we can say that in the APW method the wave function is approximated in the interstitial region by plane waves, whereas in the core region the rapid oscillations are automatically incorporated via direct integration of the Schrödinger equation. The basis functions are continuous at the sphere boundaries, but their derivative is not. The APW functions are not exact solutions to the Schrödinger equation, but they are appropriate basis functions for expanding the actual wave function:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \sum_{\mathbf{K}} C_{\mathbf{K}} \psi^{\mathrm{APW}}_{\mathbf{k}+\mathbf{K}}(\mathbf{r}). \tag{6.31}$$

The muffin tin parts of the ψ_{APW} in this expansion are all evaluated at the same energy E. The coefficients $C_{\mathbf{K}}$ are given by the lowest energy solution of the generalised eigenvalue equation:

$$\mathbf{HC} = E\mathbf{SC} \tag{6.32}$$

where the matrix elements of $\mathbf{H}$ and $\mathbf{S}$ are given by quite complicated expressions. In the resulting solution, the mismatch in the derivative across the sphere boundary is minimised.

Before giving the matrix elements of the Hamiltonian and the overlap matrix, we must point out that (6.32) differs from usual generalised eigenvalue equations in that the matrix elements of the Hamiltonian *depend on energy*. This dependence is caused by the fact that they are calculated as matrix elements of energy-dependent wave functions (remember the radial wave functions depend on the energy). Straightforward application of the matrix methods for generalised eigenvalue problems is therefore impossible.

In order to obtain the spectrum, we rewrite Eq. (6.32) in the following form:

$$(\mathcal{H} - E)\mathbf{C} = 0 \tag{6.33}$$

where $\mathcal{H} = \mathbf{H} - E\mathbf{S} + E\mathbf{I}$ ($\mathbf{I}$ is the unit matrix). Although the form (6.33) suggests that we are dealing with an ordinary eigenvalue problem, this is not the case: the overlap matrix has been moved into $\mathcal{H}$. To find the eigenvalues we calculate the determinant $|\mathcal{H} - E\mathbf{I}|$ on a fine energy mesh and see where the zeroes are. It is sometimes possible to use root-finding algorithms (see Appendix A3) to locate the zeroes of the determinant, but these often fail because the energy levels may be calculated along symmetry lines in the Brillouin zone and in that case the determinant may vary quadratically about the zero, so that it becomes impossible to locate this point by detecting a change of sign (this problem does not arise in Green's function approaches, where one evaluates the Green's function at a definite energy).

For crystals with one atom per unit cell, the matrix elements of the Hamiltonian for APW basis functions with wave vectors $\mathbf{q}_i = \mathbf{k} + \mathbf{K}_i$ and $\mathbf{q}_j = \mathbf{k} + \mathbf{K}_j$ are given by

$$\mathcal{H}_{ij} = \langle \mathbf{k} + \mathbf{K}_i | \mathcal{H} | \mathbf{k} + \mathbf{K}_j \rangle = -EA_{ij} + B_{ij} + \sum_{l=0}^{l_{\max}} C_{ijl} \frac{\mathcal{R}'_l(R)}{\mathcal{R}_l(R)}. \tag{6.34}$$

In this expression, $\mathcal{R}'_l(R)$ is $d\mathcal{R}_l(r)/dr$, evaluated at the muffin tin boundary $r = R$. The coefficients A_{ij}, B_{ij} and C_{ijl} are given by

$$\begin{aligned} A_{ij} &= \frac{-4\pi R^2}{\Omega} \frac{j_1(|\mathbf{K}_i - \mathbf{K}_j|R)}{|\mathbf{K}_i - \mathbf{K}_j|} + \delta_{ij}, \\ B_{ij} &= \frac{1}{2} A_{ij} (\mathbf{q}_i \cdot \mathbf{q}_j) \text{ and} \\ C_{ijl} &= (2l+1) \frac{2\pi R^2}{\Omega} P_l \left(\frac{\mathbf{q}_i \cdot \mathbf{q}_j}{q_i q_j} \right) j_l(q_i R) j_l(q_j R). \end{aligned} \tag{6.35}$$

Here, Ω is the volume of the unit cell. Note that there is no divergence in the expression for the matrix elements A_{ij} if $|\mathbf{K}_i - \mathbf{K}_j|$ vanishes, as $j_1(x) \to x/3$ for small x.

In addition to the inconvenient energy dependence of the Hamiltonian, another problem arises from the occurrence of $\mathcal{R}_l(R)$ in the denominator in (6.34). For energies for which the radial solution $\mathcal{R}_l$ happens to be zero or nearly zero on the border of the muffin tin spheres, the matrix elements become very large or even diverge, which may cause numerical instabilities. In the linearised APW (LAPW) method, an energy-independent Hamiltonian is used in which the radial solution does not occur in a denominator, and therefore both problems of the APW method are avoided. The APW method is hardly used now because of the energy-dependence problem. The reasons we have treated it here are that it is conceptually simple and that the principle of this method lies at the basis of many other methods. Further details on the APW method can be found in Refs. [14–16].

6.5.2 An APW program for the band structure of copper

Copper has atomic number $Z = 29$. We consider only the valence states. The core states are two s- and two p-states, and theeleven valence electrons occupy the third s-level and the first d-levels. Its crystal structure is fcc (Figure 6.1) with lattice constant $a = 6.822$ a.u. The unit cell volume Ω is equal to $3a^3/4$. The reciprocal lattice is a bcc lattice with basis vectors

$$\mathbf{b_1} = \frac{2\pi}{a} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b_2} = \frac{2\pi}{a} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{b_3} = \frac{2\pi}{a} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}. \tag{6.36}$$

The Brillouin zone of this lattice is shown in Figure 6.9.

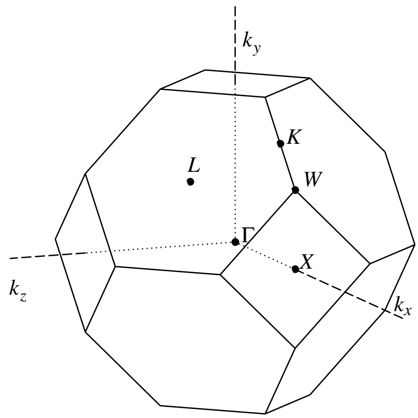


Figure 6.9. Brillouin zone of the fcc lattice.

For a given vector $\mathbf{k}$ in the Brillouin zone, we construct the APW basis vectors as $\mathbf{q} = \mathbf{k} + \mathbf{K}$. The norm of a reciprocal lattice vector $\mathbf{K} = l\mathbf{b}_1 + m\mathbf{b}_2 + n\mathbf{b}_3$ is given by

$$|\mathbf{K}| = \frac{2\pi}{a}\sqrt{3l^2 + 3m^2 + 3n^2 - 2lm - 2nl - 2nm}. \tag{6.37}$$

We take a set of reciprocal lattice vectors with norm smaller than some cut-off and it turns out that the sizes of these sets are 1, 9, 15, 27 etc. A good basis set size to start with is 27, but you might eventually do calculations using 113 basis vectors, for example. The set of such reciprocal lattice vectors is easy to generate by considering all vectors with l, m and n between say -6 and 6 and neglecting all those with norm beyond some cut-off.

The program must contain loops over sets of k-points in the Brillouin zone between for example Γ and X in Figure 6.9. The locations of the various points indicated in Figure 6.9, expressed in cartesian coordinates, are

$$\Gamma = \frac{2\pi}{a}\begin{pmatrix}0\\0\\0\end{pmatrix}, \quad X = \frac{2\pi}{a}\begin{pmatrix}1\\0\\0\end{pmatrix}, \quad K = \frac{2\pi}{a}\begin{pmatrix}3/4\\3/4\\0\end{pmatrix},$$
$$W = \frac{2\pi}{a}\begin{pmatrix}1\\1/2\\0\end{pmatrix}, \quad L = \frac{2\pi}{a}\begin{pmatrix}1/2\\1/2\\1/2\end{pmatrix}. \tag{6.38}$$

For each $\mathbf{k}$, the matrix elements A_{ij}, B_{ij} and C_{ijl} in (6.35) are to be determined. Good values for the cut-off angular momentum are $l_{\text{max}} = 3$ or 4. Then, for any energy E, the matrix elements of $\mathcal{H}$ according to (6.34) can be found by first solving the radial Schrödinger equation numerically from $r = 0$ to $r = R$ and then using the

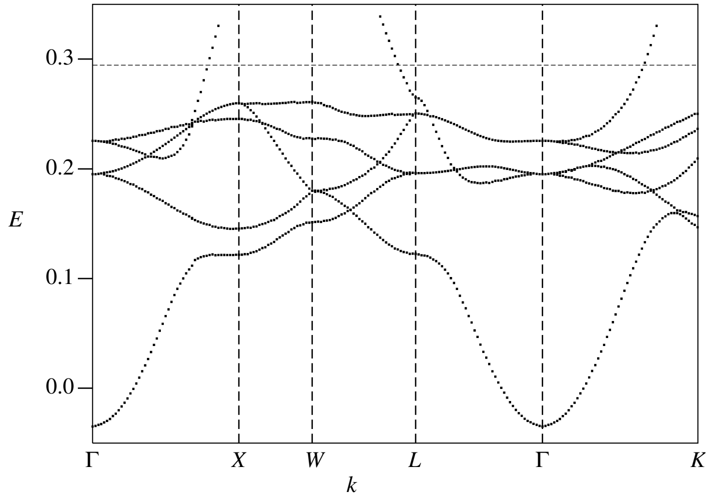


Figure 6.10. Band structure of fcc copper. The Fermi energy is shown as a horizontal dashed line.

quotient $\mathcal{R}_l'(R)/\mathcal{R}_l(R)$ as obtained from this solution in (6.34). Our program will not be self-consistent as we shall use a reasonable one-electron potential.[2]

It is best to use some numerical routine for calculating the determinant. If such a routine is not available, you can bring your matrix to an upper-triangular form as described in Appendix A8 and multiply the diagonal elements of the resulting upper triangular matrix to obtain the determinant.

If you have a routine at your disposal which can calculate the determinant for each k-vector and for any energy, the last step is to calculate the eigenvalues (energies) at some **k**-point. This is a very difficult step and you are advised not to put too much effort into finding an optimal solution for this. The problem is that often the determinant may not change sign at a doubly degenerate level, and energy levels may be extremely close. Finally, changes of sign may occur across a singularity. A highly inefficient but fool-proof method is to calculate the determinant for a large amount of closely spaced energies containing the relevant part of the spectrum (to be read off from Figure 6.10) and then scan the results for changes of sign or near-zeroes. It is certainly not advisable to try writing a routine which finds all the energy eigenvalues automatically using clever root-finding algorithms.

[2] The potential can be found on www.cambridge.org/9780521833469. There are in fact two files in which the potential is given on different grids: the first one is a uniform grid and the second an exponential grid considered in Problem 5.1. Details concerning the integration of the Schrödinger equation on the latter are to be found in this problem.

PROGRAMMING EXERCISE

Write a program for calculating the determinant $|\mathcal{H} - E|$.

Check Check that the determinant vanishes near the values which you can read off from Figure 6.10 for a few points in the Brillouin zone.

The Fermi level for the potential supplied lies approximately at 0.29 a.u., so one conclusion you can draw from the resulting band structure is that copper is a conductor as the Fermi energy does not lie in the energy gap.

You will by now have appreciated why people have tried to avoid energy-dependent Hamiltonians. In the next section we shall describe the linearised APW (LAPW) method which is based on the APW method, but avoids the problems associated with the latter.

6.6 The linearised APW (LAPW) method

A naive way of avoiding the energy-dependence problem in APW calculations would be to use a fixed 'pivot' energy for which the basis functions are calculated and to use these for a range of energies around the pivot energy. If the form of the basis functions inside the muffin tin varies rapidly with energy (and this turns out to be often the case) this will lead to unsatisfactory results.

The idea of the LAPW method [17, 18] is to use a set of pivot energies for which not only the solution to the radial Schrödinger equation is taken into account in constructing the basis set, but also its energy derivative. This means that the new basis set should be adequate for a range of energies around the pivot energy in which the radial basis functions can be reasonably approximated by an energy linearisation:

$$\mathcal{R}(r, E) = \mathcal{R}(r, E_p) + (E - E_p)\dot{\mathcal{R}}(r, E_p). \tag{6.39}$$

Here, and in the remainder of this section, the dot stands for the energy derivative, as opposed to the prime, which is used for the radial derivative – for any differentiable function $f(r, E)$:

$$\dot{f}(r, E) = \frac{\partial}{\partial E} f(r, E) \quad \text{and} \tag{6.40a}$$

$$f'(r, E) = \frac{\partial}{\partial r} f(r, E). \tag{6.40b}$$

The energy derivatives of the radial solution within the muffin tins are used alongside the radial solutions themselves to match onto the plane wave outside the spheres. Note that the APW Hamiltonian depends on energy only via the radial solutions $\mathcal{R}_l$, so if we take these solutions and their energy derivatives $\dot{\mathcal{R}}_l$ at a *fixed* energy into account, we have eliminated all energy dependence from the Hamiltonian.

In comparison with the APW method, we have twice as many radial functions inside the muffin tin sphere, $\mathcal{R}_l$ and $\dot{\mathcal{R}}_l$, and we can match not only the value but also the derivative of the plane wave $\exp(\mathrm{i}\mathbf{q}\cdot\mathbf{r})$ across the sphere boundary. We write the wave function inside as the expansion

$$\Phi_{\mathbf{k}+\mathbf{K}}(\mathbf{r}) = \sum_{lm}[A_{lm}\mathcal{R}_l(r;E_p) + B_{lm}\dot{\mathcal{R}}_l(r;E_p)]Y^l_m(\theta,\phi) \tag{6.41}$$

and the numbers A_{lm} and B_{lm} are fixed by the matching condition. There is no energy dependence of the wave functions, and they are smooth across the sphere boundary, but the price which is paid for this is giving up the exactness of the solution inside the sphere for the range of energies we consider.

We end up with a generalised eigenvalue problem with energy-independent overlap and Hamiltonian matrices. These matrices are reliable for energies in some range around the pivot energy. It turns out that the resulting wave functions have an inaccuracy of $(E-E_p)^2$ as a result of the linearisation and that the energy eigenvalues deviate as $(E-E_p)^4$ from those evaluated at the correct energy – see Ref. [18].

The expressions for the matrix elements are again quite complicated. They depend on the normalisations for $\mathcal{R}_l$ and $\dot{\mathcal{R}}_l$ which will be specified below. For the coefficients A_{lm} and B_{lm}, the matching conditions lead (with $\mathbf{q} = \mathbf{k}+\mathbf{K}$, $\mathbf{q}' = \mathbf{k}+\mathbf{K}'$) to:

$$A_{lm}(\mathbf{q}) = 4\pi R^2 \mathrm{i}^l \Omega^{-1/2} Y^{l\,*}_m(\theta_q,\phi_q)a_l; \tag{6.42a}$$

$$a_l = j'_l(qR)\dot{\mathcal{R}}_l(R) - j_l(qR)\dot{\mathcal{R}}'_l(R); \tag{6.42b}$$

$$B_{lm}(\mathbf{q}) = 4\pi R^2 \mathrm{i}^l \Omega^{-1/2} Y^{l\,*}_m(\theta_q,\phi_q)b_l; \tag{6.42c}$$

$$b_l = j_l(qR)\mathcal{R}'{}_l(R) - j'_l(qR)\mathcal{R}_l(R). \tag{6.42d}$$

The matrix elements of the overlap matrix and the Hamiltonian can now be calculated straightforwardly – the result for the overlap matrix is [18]

$$S_{\mathbf{K},\mathbf{K}'} = U(\mathbf{K}-\mathbf{K}') + \frac{4\pi R^4}{\Omega}\sum_l (2l+1)P_l(\hat{\mathbf{q}}\cdot\hat{\mathbf{q}}')s^l_{\mathbf{K},\mathbf{K}'} \quad \text{with} \tag{6.43a}$$

$$s^l_{\mathbf{K},\mathbf{K}'} = a_l(q)a_l(q') + b_l(q)b_l(q')N_l \quad \text{and} \tag{6.43b}$$

$$U(\mathbf{K}) = \delta_{\mathbf{K},\mathbf{O}} - \frac{4\pi R^2}{\Omega}\frac{j_l(KR)}{K}. \tag{6.43c}$$

Here, N_l is the norm of the energy derivative inside the muffin tin (see below). The Hamiltonian is given by

$$H_{\mathbf{K},\mathbf{K}'} = (\mathbf{q}\cdot\mathbf{q}')U(\mathbf{K}-\mathbf{K}') + \frac{4\pi R^2}{\Omega}\sum_l (2l+1)P_l(E_l s^l_{\mathbf{K},\mathbf{K}'} + \gamma_l) \tag{6.44a}$$

with

$$\gamma_l = \mathcal{R}'_l(R)\dot{\mathcal{R}}_l(R)[j_l'(qR)j_l(q'R) + j_l(qR)j_l'(q'R)] \\ - [\mathcal{R}'_l(R)\dot{\mathcal{R}}_l'(R)j_l(qR)j_l(q'R) + \mathcal{R}_l(R)\dot{\mathcal{R}}_l(R)j_l'(qR)j_l'(q'R)]. \tag{6.44b}$$

We see that a pleasing feature of these expressions is that we do not get APW-type numerical inaccuracies due to radial solutions vanishing at the muffin tin radius and occurring in the denominator of the expressions for the matrix elements.

Finally, we must find out how the energy derivative of the solution of the radial Schrödinger equation, $\dot{\mathcal{R}}_l$, can be calculated. By differentiating the radial Schrödinger equation

$$(H - E)\mathcal{R}_l(r; E) = 0 \tag{6.45}$$

with respect to E, we find that $\dot{\mathcal{R}}_l$ satisfies the following differential equation:

$$(H - E)\dot{\mathcal{R}}_l(r; E) = \mathcal{R}_l(r; E). \tag{6.46}$$

This second order inhomogeneous differential equation needs two conditions to fix the solution. The first condition is that $\dot{\mathcal{R}}_l$ (like $\mathcal{R}_l$) is regular at the origin which leaves the freedom of adding $\alpha\mathcal{R}_l(r)$ to it, for arbitrary α ($\mathcal{R}_l$ is the solution of the homogeneous equation). The number α is fixed by the requirement that $\mathcal{R}_l$ is normalised:

$$\int_0^R \mathrm{d}r\, r^2\mathcal{R}_l^2(r; E) = 1 \tag{6.47}$$

which, after differentiation with respect to E, leads to

$$\int_0^R r^2\mathcal{R}_l(r)\dot{\mathcal{R}}_l(r)\mathrm{d}r = 0 \tag{6.48}$$

i.e. $\mathcal{R}_l$ and $\dot{\mathcal{R}}_l$ are orthogonal. The norm of $\dot{\mathcal{R}}_l$,

$$N_l = \int_0^R \mathrm{d}r r^2|\dot{\mathcal{R}}_l(r)|^2, \tag{6.49}$$

which occurs in the definition of the overlap matrix, is therefore in general not equal to one. It can be shown that the normalisation condition (6.47) leads to the following boundary condition at the muffin tin bounday ($r = R$):

$$R^2[\mathcal{R}'_l(R)\dot{\mathcal{R}}_l(R) - \mathcal{R}(R)\dot{\mathcal{R}}_l'(R)] = 1; \tag{6.50}$$

see Problem 6.5.

The interested reader might try to write a program for calculating the band structure of copper using this linearisation technique. The determination of the eigenvalues will be found much more easily than in the case of the APW calculation, as the Hamiltonian is energy-independent.

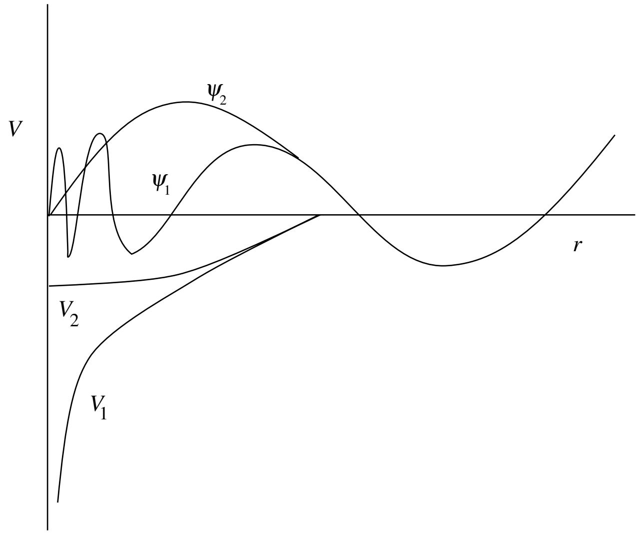


Figure 6.11. The principle of the pseudopotential. The wave functions of the full potential (Ψ_1) and of the pseudopotential (Ψ_2) are equal beyond some radius.

6.7 The pseudopotential method

We have already seen that the main problem in calculating band structures is the deep Coulomb potential giving rise to rapid oscillations close to the nuclei. In the pseudopotential method this problem is cleverly transformed away by choosing a potential which is weak. It is not immediately obvious that this is possible: after all, the solutions of the problem with a weak and with a deep potential can hardly describe the same system! The point is that the pseudopotential does not aim at describing accurately what happens in the core region, but it focuses on the valence region. A weak potential might give results that outside the core region are the same as those of the full potential.

In order to obtain a better understanding of this, we must return to Chapter 2, where the concept of phase shift was discussed. The phase shift uniquely determines the scattering properties of a potential – indeed, we seek a weak pseudopotential that scatters the valence electrons in the same way as the full potential, so that the solution beyond the core region is the same for both potentials. An important point is that we can add an integer times π to the phase shift without changing the solution outside the core region, and there exist therefore many different potentials yielding the same valence wave function. To put it another way: if we make the potential within the core region deeper and deeper, the phase shift will increase steadily, but an increase by π does not affect the solution outside. The pseudopotential is a weak potential which gives the same phase shift (modulo π) as the full potential and hence the same solution outside the core region. The principle is shown in

Figure 6.11 which shows two different potentials and their solutions (for the same energy). These solutions differ strongly within the core region but they coincide in the valence region.

What the pseudopotential does is to remove nodes from the core region of the valence wave function while leaving it unchanged in the valence region. The nodes in the core region are necessary in order to make the valence wave functions orthogonal to the core states. If there are no core states for a given l, the valence wave function is nodeless and the pseudopotential method is less effective. Such is the case in 3d transition metals, such as copper.

The phase shift depends on the angular momentum l and on the energy. A pseudopotential that gives the correct phase shift will therefore also depend on these quantities. The energy dependence is particularly inconvenient, as we have seen in the discussion of the APW method. In Section 6.7.2 we shall see that this dependence disappears automatically when solving another problem associated with the pseudopotential: that of the distribution of the charge inside and outside the core region. More details will be given in that section, and we restrict ourselves here to energy-independent pseudopotentials.

There is a considerable freedom in choosing the pseudopotential as it only has to yield the correct phase shift outside the core region, and several simple parametrised forms of pseudopotentials have been proposed. These are fitted either to experimental data for the material in question (the *semi-empirical* approach), or to data obtained using *ab initio* methods for ions and atoms of the same material, obtained using full-potential calculations.

We give two examples of pseudopotentials.

- The Ashcroft empty-core pseudopotential [19]:

$$V(r) = \begin{cases} -\dfrac{Ze}{r}, & r > r_c \\ 0 & r < r_c \end{cases} . \tag{6.51}$$

  Z is the valence of the ion and there is only one parameter to be adjusted: the cut-off length r_c. Although its simplictity is very attractive, this potential does not perform very well for wide energy ranges, although it reproduces some material properties reasonably well.
- The Fourier-component parametrisation

$$V(r) = \sum\nolimits'_{\mathbf{K}} V_{\mathbf{K}} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{R}}. \tag{6.52}$$

  where the sum $\sum'$ is over a limited set of $\mathbf{K}$-vectors. This parametrisation is convenient for the plane wave basis set which is (nearly) always used in pseudopotential calculations. In the next subsection we shall use this form of the pseudopotential in a band structure program for silicon.

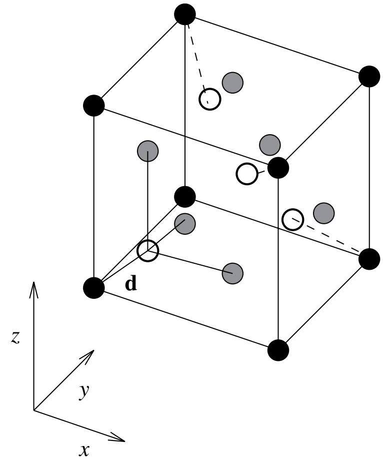


Figure 6.12. The diamond structure.

There are numerous review articles on the pseudopotenial method and readers who are interested in the subject are referred to those by Heine [10], Brust [20] and Pickett [21].

6.7.1 A pseudopotential band structure program for silicon

In this section, the construction of a pseudopotential program for silicon is described. For details, the review by Brust [20] and the paper by Chelikowsky and Cohen [22] may be consulted.

Silicon is considered here in the diamond structure which is a fcc lattice with, at each lattice point, two atoms at relative positions $\pm 1/8(\mathbf{a_1} + \mathbf{a_2} + \mathbf{a_3})$ (see Figure 6.12). We have already described the fcc crystal structure and the special points in the first Brillouin zone in Section 6.5. The lattice constant is $5.43a_0$. The pseudopotential is given in the convenient form of a few Fourier components (see above). We restrict the number of coefficients by assuming the pseudopotential to be a repetition of spherically symmetric potentials in cells surrounding the atoms, which leads to the following form of the Fourier components of the pseudopotential $V_{\rm ps}$ arising from a single atom per cell:

$$V_{\rm ps}^{\rm (at)}(K) = \frac{1}{V_{\rm cell}} \int_{\rm cell} \mathrm{d}^3r \, V_{\rm ps}^{\rm (at)}(r) \mathrm{e}^{-\mathrm{i}\mathbf{K}\cdot\mathbf{r}}. \tag{6.53}$$

The Fourier components depend only on the length of the wave vector **K**, and this property reduces the number of independent Fourier coefficients.

Another reduction comes about when calculating the Fourier component of the sum of the potentials arising from the two atoms at positions

$\mathbf{d}_{1,2} = \pm 1/8(\mathbf{a_1} + \mathbf{a_2} + \mathbf{a_3})$ relative to the lattice points:

$$V_{\rm ps}^{\rm (tot)}(\mathbf{K}) = V_{\rm ps}^{\rm (at)}(K)(\mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{d}_1} + \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{d}_2}). \tag{6.54}$$

The sum of the exponentials on the right hand side is known as the *structure factor*. Therefore, we find for a vector $\mathbf{K} = \sum_{i=1}^{3} n_i \mathbf{b}_i$ that

$$V_{\rm ps}^{\rm (tot)}(\mathbf{K}) = \cos[\pi(n_1 + n_2 + n_3)/4] V_{\rm ps}^{\rm (at)}(K). \tag{6.55}$$

It follows immediately that the pseudopotential components vanish if the sum of the n_i is an odd multiple of 2: the structure factor causes extinction of certain wave vectors. Furthermore, we can choose the component $V_{\rm ps}^{\rm (tot)}(\mathbf{K} = \mathbf{0})$ to be equal to zero, as this induces a mere shift in the energy offset. Collecting all bits and pieces, we are left with the following values of $|\mathbf{K}| = K$ for which the pseudopotential does not vanish (apart from a factor $4\pi^2/a^2$):

$$K^2 = 3, 8, 11, \ldots \tag{6.56}$$

and only these first three components are taken into account in the pseudopotential. This means that only three numbers are to be fitted and the whole band structure follows from them.

We shall not carry out the fitting procedure but quote the resulting values for the potential from literature, resulting from a fit to optical transition energies [22] – they read (in atomic units):

$$V_{\rm ps}^{\rm (tot)}(\sqrt{3}) = -0.1121; \; V_{\rm ps}^{\rm (tot)}(\sqrt{8}) = 0.0276; \; V_{\rm ps}^{\rm (tot)}(\sqrt{11}) = 0.0362. \tag{6.57}$$

The matrix element of the pseudopotential Hamiltonian for plane waves $\mathbf{k} + \mathbf{K}$ and $\mathbf{k} + \mathbf{K}'$ is given by $(\Delta\mathbf{K} = \mathbf{K} - \mathbf{K}')$

$$H_{\mathbf{K},\mathbf{K}'} = \frac{1}{2}|\mathbf{k} + \mathbf{K}|^2 \delta_{\mathbf{K},\mathbf{K}'} + V(|\Delta\mathbf{K}|) \cos\left[(\Delta K_1 + \Delta K_2 + \Delta K_3)\frac{\pi}{4}\right]. \tag{6.58}$$

The diagonalisation of the resulting matrix is straightforward: the plane waves are orthogonal and hence no overlap matrix has to be taken into account. You can use basis sets of size 9, 15, 27 etc., just as in the APW case. In Figure 6.13 the band structure is represented for a basis with 113 states.

The band structure in Figure 6.13 matches the results of calculations using more sophisticated methods very well, which is remarkable if you note that only three free parameters enter into the potential. The fact that the band gap comes out well is not surprising since it has been used in the fitting procedure. It turns out to be 1.17 eV, and you might compare this with $k_{\rm B}T$ at room temperature in order to estimate the fraction of electrons excited into the conduction band using the Fermi–Dirac distribution.

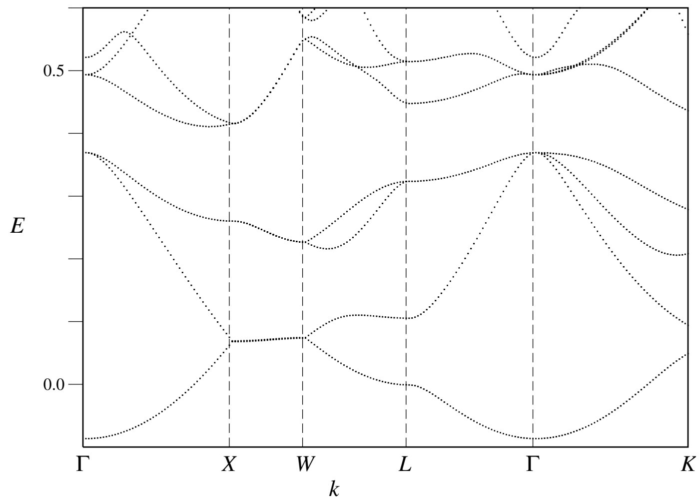


Figure 6.13. Band structure of silicon.

6.7.2 Accurate energy-independent pseudopotentials

Suppose we have a pseudopotential that gives exactly the same phase shift as the full potential. In the valence region, the wave functions have the same shape as for the full potential, but their normalisation may differ: the wave functions in the valence region for the two potentials are the same only up to a scaling factor. The point is that if two normalised wave functions differ within the core region while being similar (up to a multiplication constant) in the valence region, their respective charges will be distributed differently among core and valence regions. The resulting charge difference is called *orthogonality hole* and one should correct for it, for example by rescaling the full pseudo-wave function.

It turns out that the normalisation of the states is related to energy dependence of the pseudopotential. It can be shown (see Problem 6.1) that for the *full* potential, the charge inside a sphere around the nucleus with radius R_c carried by the solution ψ of the Schrödinger equation evaluated at energy E is related to the energy derivative of the wave function at R_c:

$$\int_{\text{core}} \mathrm{d}^3 r |\psi(\mathbf{r})|^2 = -\frac{1}{2} \int \mathrm{d}\Omega\, R_c^2\, \psi(\mathbf{r}) \frac{\partial^2 \psi(\mathbf{r})}{\partial r \partial E}\bigg|_{r=R_c} \tag{6.59}$$

where the integration is carried out over the spherical angles, $\mathrm{d}\Omega = \mathrm{d}\cos\vartheta\, \mathrm{d}\varphi$. This is another instance of the relation between norm and energy derivative which was previously encountered in connection with the energy derivatives of the solution of the radial equation in the LAPW wave functions in Section 6.6.

If we now consider an energy-dependent pseudopotential $V_{\rm ps}$ with eigenstate ϕ, we obtain, aside from the surface integral on the right hand side, an integral over the energy derivative of $V_{\rm ps}$:

$$\int_{\rm core} \mathrm{d}^3r|\phi(\mathbf{r})|^2 = -\frac{1}{2}\int \mathrm{d}\Omega R_{\rm c}^2\, \phi(\mathbf{r})\frac{\partial^2\phi(\mathbf{r})}{\partial r\partial E}\bigg|_{r=R_{\rm c}} + \int_{\rm core} \mathrm{d}^3r\frac{\partial V(\mathbf{r},E)}{\partial E}|\phi(\mathbf{r})|^2. \tag{6.60}$$

The first term on the right hand side is equal to the right hand side in (6.59) if we fix the amplitude of ϕ to be equal to that of ψ at $R_{\rm c}$. Therefore, if both solutions have the same amount of charge inside the core region, the second term on the right hand side must vanish, which implies that the pseudopotential is independent of energy. This means that if we have solved the orthogonality hole problem, we have obtained an energy-independent pseudopotential, so that we have solved two problems at once.

Bachelet, Hamann and Schlüter [23] have constructed accurate norm-conserving pseudopotentials and we refer to their paper and to the review article by Pickett [21] for further details. Goedecker, Teter and Hutter [24] have developed a particularly convenient type of pseudopotential which, being based on Gaussian functions, allows for analytic Fourier transforms. We shall use this pseudopotential in the following section in the construction of a fully self-consistent pseudopotential program.

6.7.3 Building a self-consistent pseudopotential program

The construction of a fully self-consistent pseudopotential program is quite elaborate – we shall therefore restrict ourselves to the case of a cubic unit cell and instead of summing over all points in the Brillouin zone, we shall only consider the Γ-point (i.e. reciprocal vector $\mathbf{K} = \mathbf{0}$). This restriction is often applied when dealing with molecules: the cell is taken big enough to ensure that the electron density vanishes near the cell boundary. This choice therefore renders our program more suitable for molecular systems or clusters than for periodic solids. However, the method for periodic systems uses very similar techniques, and the interested reader is invited to extend his or her program to that case.

It is important to build up the program in a step by step fashion and check each step very carefully. The steps are described below. We closely follow the setup of the CPMD program, described in the review paper by Marx and Hutter [25]. For more details concerning the program and further background, that paper should be consulted.

We start with some remarks concerning definitions and conventions relating to Fourier transforms and choice of basis functions. For simplicity, we restrict ourselves to cubic unit cells. Although all quantities are expanded in a basis of plane waves with wave vectors on a grid in reciprocal space, we cannot always use periodicity on this grid. Consider for example the potential of a nucleus or ion core, located at position $\mathbf{r}_0$ in the unit cell. We expand this potential using a grid of wave vectors

$$\mathbf{K} = \frac{2\pi}{L}(n_x, n_y, n_z) \tag{6.61}$$

with n_x running from 1 to N, L/N being the resolution in real space. If we represent the potential in *real* space on the real-space grid, its Fourier transform is periodic in K-space, with a period $2\pi N/L$ in each Cartesian direction. Usually we are given the (Fourier transform of the) pseudopotential for an ion core located at the origin. If the atom is actually located at some place $\mathbf{r}_0$, the Fourier transfrom acquires an extra structure factor $\exp(-\mathrm{i}\mathbf{K}\cdot\mathbf{r}_0)$. If $\mathbf{r}_0$ does not lie on the real-space grid, this structure factor is nonperiodic in reciprocal space!

Another example of a nonperiodic operator in reciprocal space is the kinetic energy, for which we know the Fourier transform of the operator in continuum space:

$$T = \frac{\hbar^2 K^2}{2m} \tag{6.62}$$

(in atomic units, this reduces to $K^2/2$). This expression is cut off at some maximum wave vector beyond which the components of the orbitals are supposed to be very small. Note that we do not use the periodic discrete form of the kinetic energy (with Fourier transform $3-\cos(K_x a)-\cos(K_y a)-\cos(K_z a)$; $a = L/N$); the form (6.61) is a more accurate representation.

As a basis, we use Fourier waves $\mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}}/\sqrt{\Omega}$ (remember that Ω is volume of the unit cell). An orbital $\phi^{(j)}$ is then expanded in these basis vectors as

$$\phi^{(j)}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}}\sum_{\mathbf{K}} c_{\mathbf{K}}^{(j)} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}}. \tag{6.63}$$

The coefficients $c_{\mathbf{K}}^{(j)}$ come out of a diagonalisation routine and are usually normalised according to

$$\sum_{\mathbf{K}} |c_{\mathbf{K}}^{(j)}|^2 = 1. \tag{6.64}$$

Therefore we have:

$$\sum_{\mathbf{r}} |\phi^{(j)}(\mathbf{r})|^2 = \frac{1}{\Omega}\sum_{\mathbf{r}}\sum_{\mathbf{K},\mathbf{K}'} c_{\mathbf{K}}^{(j)} c_{\mathbf{K}'}^{(j)*} \mathrm{e}^{\mathrm{i}(\mathbf{K}-\mathbf{K}')\cdot\mathbf{r}} = \frac{N^3}{\Omega}. \tag{6.65}$$

The density due to all orbitals is therefore given by

$$n(\mathbf{r}) = \sum_j f_j |\phi^{(j)}(\mathbf{r})|^2, \tag{6.66}$$

where f_j is the *occupancy*, which is usually the Fermi–Dirac function with an additional factor of 2 in the closed-shell case. To check that the normalisation is indeed correct, we calculate the total charge for the closed-shell system at $T = 0$:

$$\int n(\mathbf{r})\, \mathrm{d}^3 r = 2 \sum_{j_{\mathrm{occ}}} |\phi^{(j)}(\mathbf{r})|^2\, \mathrm{d}^3 r = 2\frac{\Omega}{N^3} \sum_{j_{\mathrm{occ}}} \sum_{\mathbf{r}} |\phi^{(j)}(\mathbf{r})|^2 = N_{\mathrm{el}} \tag{6.67}$$

where N_{el} is the number of electrons (the factor 2 in front of the second and third expressions is due to the spin degeneracy). Note that the prefactor Ω/N^3 results from the transition of the integral to a sum.

We can formulate Fourier transforms *in continuous, real space* by writing operators and vectors with respect to the basis functions $\exp(\mathrm{i}\mathbf{K}\cdot\mathbf{r})/\sqrt{\Omega}$. In that case, the discrete representation converges to the continuum one for fine grids, and there is no ambiguity concerning the prefactors in the Fourier transforms (powers of Ω). The only exception is the density, which is not a vector or operator in Hilbert space, and we have therefore some freedom in the definition of its Fourier transform. We adopt the convention usually taken in this field, writing

$$n(\mathbf{r}) = \sum_{\mathbf{K}} n(\mathbf{K}) \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}}. \tag{6.68}$$

It then follows from Eqs. (6.63) and (6.66) that

$$n(\mathbf{K}) = \frac{1}{\Omega} \int n(\mathbf{r}) \mathrm{e}^{-\mathrm{i}\mathbf{K}\cdot\mathbf{r}}\, \mathrm{d}^3 r = \frac{1}{\Omega} \sum_j f_j \sum_{\mathbf{K}'} c^{(j)}_{\mathbf{K}+\mathbf{K}'} c^{*(j)}_{\mathbf{K}'}. \tag{6.69}$$

An important issue concerns the truncation of the sums over $\mathbf{K}$. The point is that the potential is expressed in terms of the *density*, and not of the wave function. Now suppose that we can safely assume that the wave function vanishes for K-vectors beyond some maximum value $K_{\max}$. In that case, from working out the density in real space,

$$\begin{aligned} n(\mathbf{r}) &= \sum_j f_j |\phi_j(\mathbf{r})|^2 = \sum_j f_j \sum_{\mathbf{K},\mathbf{K}'} c^{(j)}_{\mathbf{K}} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}} c^{*(j)}_{\mathbf{K}'} \mathrm{e}^{-\mathrm{i}\mathbf{K}'\cdot\mathbf{r}} \\ &= \frac{1}{\Omega} \sum_j f_j \sum_{\mathbf{K},\mathbf{K}'} c^{(j)}_{\mathbf{K}+\mathbf{K}'} c^{*(j)}_{\mathbf{K}'} \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}} = \sum_{\mathbf{K}} n(\mathbf{K}) \mathrm{e}^{\mathrm{i}\mathbf{K}\cdot\mathbf{r}}, \end{aligned} \tag{6.70}$$

we see that $n(\mathbf{K})$ contains contributions $\mathbf{K} - \mathbf{K}'$ running up to $2K_{\max}$! Therefore, the potential also contains nonzero components for K up to $2K_{\max}$. To see that these terms occur in the Hamiltonian matrix, we consider now a *local* potential: this is a potential which depends only on $\mathbf{r}$. Fourier transforming leads to a potential $V_{\mathbf{K},\mathbf{K}'}$ in reciprocal space which is translationally invariant:

$$V_{\mathbf{K},\mathbf{K}'} = V(\mathbf{K} - \mathbf{K}'). \tag{6.71}$$

Even if both $|\mathbf{K}|$ and $|\mathbf{K}'|$ are smaller than $K_{\max}$, their difference can attain lengths up to $2K_{\max}$!

A two-dimensional representation of the situation is depicted in Figure 6.14, where the sphere of radius $K_{\max}$ is indicated as a dashed circle, called $C1$. The kinetic energy is evaluated only for values of $\mathbf{K}$ within this circle. When evaluating the kinetic energy, only the points lying inside $C1$ must be taken into account. However, the Fourier transorm of the density, which satisfies periodicity in K-space, has nonzero components for points inside the bigger circle $C2$.

Now let us again consider the contribution to the Hamiltonian of a *local potential*. The matrix elements are given by

$$V_{\mathbf{K},\mathbf{K}'} = \frac{1}{\Omega}\int \mathrm{e}^{-\mathrm{i}(\mathbf{K}-\mathbf{K}')\cdot\mathbf{r}} V(\mathbf{r})\,\mathrm{d}^3r \approx \frac{1}{N^3}\sum_{\mathbf{r}} \mathrm{e}^{-\mathrm{i}(\mathbf{K}-\mathbf{K}')\cdot\mathbf{r}} V(\mathbf{r}) = V(\mathbf{K}-\mathbf{K}'). \tag{6.72}$$

The last expression on the right hand side is the discrete Fourier transform of the potential in real space. We regularly must perform Fourier transforms, for which we use the FFT algorithm (see Appendix A9). There exist packages containing FFT routines, and we mention here the FFTW package (`http:// www.fftw.org`). Often, these packages have their own type definitions for real and complex numbers, which should then be used throughout your program.

The simplest possible nontrivial case is the one with seven wave vectors in $C1$: one in the origin and two along each of the three Cartesian axes. These are used for the Hamiltonian and the wave functions. The wave vectors for which the density is evaluated run over a grid with a linear size at least four times as large as the cut-off wave vector $K_{\max}$ (see Figure 6.14), which would suggest a unit cell with a side of four grid points. We take the grid size one larger (i.e. a $5 \times 5 \times 5$ grid) in order to avoid the coincidence of point pairs like $(2, 0, 0)$ and $(-2, 0, 0)$ for functions or operators which are nonperiodic in reciprocal space (such as the pseudopotential of an ion which is not located at a real-space grid point, see above).

6.7.4 *Free particle in a box*

We start by considering a free particle in the box. The Hamiltonian only contains the kinetic term:

$$H_{\mathbf{K},\mathbf{K}'} = \frac{K^2}{2}\delta(\mathbf{K}-\mathbf{K}'). \tag{6.73}$$

For a box of size $L \times L \times L$, the seven vectors $\mathbf{K}$ we take into account are the null vector and the vectors with size $2\pi/L$ along the positive and negative Cartesian axes. The eigenvalues are therefore equal to 0 (with multiplicity 1) and $2\pi^2/L^2$ (with multiplicity 6). If we put four electrons in the box, the total energy is given by

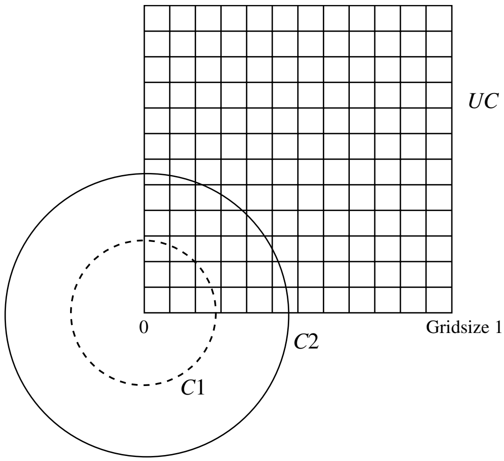


Figure 6.14. Two-dimensional representation of the reciprocal grid. The dashed sphere $C1$ contains the wave vectors of the wavefunctions – its periodic images are also shown as solid circles. The bigger circle $C2$ contains the reciprocal wave vectors for representing the electron density, and we need all those points for accurately constructing the Hamiltonian. The cell UC is a unit cell of the reciprocal lattice.

$4\pi^2/L^2$, as two of these have energy 0 and the other two are divided over the second level. In fact, in each of the six degenerate levels, we should put 1/3 electron because of symmetry. This is an example of fractional filling resulting from degeneracy.

PROGRAMMING EXERCISE

Write a program which diagonalises the Hamiltonian for a particle confined to a box.

Check For the density in a box of size 5 a.u. we find in this case that the density is homogeneous and equals $0.032 = 4/125$. This is not surprising as we put four electrons in a box of volume $5^3 = 125$.

6.7.5 Adding a pseudopotential

The pseudopotential is part of the total potential felt by the electrons. The pseudopotential consists of a local and a nonlocal part. A *local* potential can be evaluated as in Eq. (6.71). The local pseudopotential potential depends only on $\mathbf{r} - \mathbf{r}_0$, where $\mathbf{r}_0$ is the centre of an atom. We have:

$$V_{\mathbf{K},\mathbf{K}'} = V(\mathbf{K} - \mathbf{K}') = \mathrm{e}^{-\mathrm{i}(\mathbf{K}-\mathbf{K}')\cdot\mathbf{r}_0} \int \mathrm{e}^{-\mathrm{i}(\mathbf{K}-\mathbf{K}')\cdot\mathbf{r}} V(\mathbf{r})\, \mathrm{d}^3 r. \tag{6.74}$$

Note that $\mathbf{K}$ and $\mathbf{K}'$ are indices of the Hamiltonian – therefore they lie inside the $C1$ in Figure 6.14. Their difference will be inside $C2$.

For the j-th component ($j = x, y, z$) of $\mathbf{K}$, $K_j = 2\pi n_j/L$, the periodic image lying on the grid in Fig. 6.14 is found as follows:

$$K_j = 2\pi(n_j \mod \text{GridSize}), \tag{6.75}$$

where [...] denotes the integer part.

For $\mathbf{r}_0$ on the real-space grid, the structure factor in front of the integral is periodic in K-space. However, an atom does not know how we define our grid and may be located in between grid points (of course, with only one atom in the cell, we could always move the atom to $\mathbf{R} = 0$). This nonperiodicity of the structure factor is responsible for the difference between potential at the points $(2, 0, 0)$ and $(-2, 0, 0)$ (in units of the reciprocal grid constant) – hence we cannot take a grid size of 4 units, but instead need a $5 \times 5 \times 5$ grid, as mentioned above.

We now give a specific form of the pseudopotential. We shall use the Goedecker–Teter–Hutter (GTH) potential described in refs. [24] and [26]. This potential for a core, located at the origin, with s and p electrons has the form:[3]

$$V(\mathbf{r}, \mathbf{r}') = V_{\text{core}}(\mathbf{r}) + V_{\text{loc}}(\mathbf{r})\delta(\mathbf{r} - \mathbf{r}') + V_{\text{nonloc}}(\mathbf{r}, \mathbf{r}') \tag{6.76}$$

with

$$V_{\text{core}} = -\frac{Z_{\text{eff}}}{r}\text{erf}\left(\frac{r}{\sqrt{2}\xi}\right); \tag{6.77a}$$

$$V_{\text{loc}}(\mathbf{r}) = \exp[-(r/\xi)^2/2] \times [C_1 + C_2(r/\xi)^2], \tag{6.77b}$$

and

$$\begin{aligned} V_{\text{nonloc}}(\mathbf{r}, \mathbf{r}') = & \sum_{i=1}^{2} Y_0^0(\hat{\mathbf{r}}) p_i^0(r) h_i^0 p_i^0(r') Y_0^{0*}(\hat{\mathbf{r}}') \\ & + \sum_{m=1,0,-1} Y_m^1(\hat{\mathbf{r}}) p_1^1(r) h_1^1 p_1^1(r') Y_m^{1*}(\hat{\mathbf{r}}') \end{aligned} \tag{6.78}$$

In these expressions, ξ, C_i, h_i^l are parameters, and the p_i^l are the functions

$$p_1^l(r) = \sqrt{2}\frac{r^l e^{-(1/2)(r/r_l)^2}}{r^{l+3/2}\sqrt{\Gamma(l + 3/2)}} \text{and} \tag{6.79}$$

$$p_2^l(r) = \sqrt{2}\frac{r^{l+2} e^{-(1/2)(r/r_l)^2}}{r^{l+7/2}\sqrt{\Gamma(l + 7/2)}} \tag{6.80}$$

[3] The form given here is somewhat simpler than the full GTH potential. For the atoms considered here, however, the present form is sufficient.

Table 6.1. *Parameters for the GTH pseudopotential*

	Hydrogen	Silicon
ξ	0.2	0.44
C_1	−4.0663326	−6.9136286
C_2	0.6778322D0	0.0
$r_{l=0}$	–	0.4243338
$h_1^{l=0}$	–	3.2081318
$h_2^{l=0}$	–	2.5888808
$r_{l=1}$	–	0.4853587
$h_2^{l=0}$	–	2.6562230

Source: [24, 26]
Only values for hydrogen and silicon are listed.

The Gamma-function in the denominators ensures proper normalisation:

$$\int p_i^l(r) p_i^l(r) r^2 \, \mathrm{d}r = 1. \tag{6.81}$$

Let us spend a few moments studying this potential. The very first term, $-Z_{\mathrm{eff}}/r\mathrm{erf}(r/2\xi)$, is the Coulomb potential of a Gaussian charge distribution with total charge Z_{eff} and width ξ: for large arguments, that is, far from the ion core, the error function erf tends to 1. The remaining terms are short-ranged and allow therefore for refinement of the shape of the radial charge distribution. The nonlocal term is, as usual, a projection onto the different l subspaces. For a complete list of pseudopotential parameters, we refer to Refs. [24] and [26]; here we give those for hydrogen and silicon – see Table 6.1.

The Fourier transform of the GTH potential can be calculated analytically, yielding the following closed forms.

$$V_{\mathrm{core}}(\mathbf{K}) = -4\pi \frac{Z_{\mathrm{eff}}}{\Omega} \frac{\mathrm{e}^{-(K\xi)^2/2}}{K^2}; \tag{6.82}$$

$$V_{\mathrm{loc}}(\mathbf{K}) = \sqrt{(2\pi)^3} \frac{\xi^3}{\Omega} \mathrm{e}^{-(K\xi)^2/2} \{C_1 + C_2[3 - (K\xi)^2]\} \tag{6.83}$$

and

$$V_{\mathrm{nonloc}}(\mathbf{K}, \mathbf{K}') = \sum_{i=1}^{2} Y_0^0(\hat{\mathbf{K}}) p_i^0(K) h_i^0 p_i^0(K') Y_0^{0*}(\hat{\mathbf{K}}') - \sum_{m=1,0,-1} Y_m^1(\hat{\mathbf{K}}) p_1^1(K) h_1^1 p_1^1(K') Y_m^{1*}(\hat{\mathbf{K}}'). \tag{6.84}$$

The projector functions have the form p_i^l:

$$p_1^0 = \frac{1}{\sqrt{\Omega}} 4 r_s \sqrt{2 r_s} \pi^{5/4} \mathrm{e}^{-(K r_s)^2/2}, \tag{6.85a}$$

$$p_2^0 = \frac{1}{\sqrt{\Omega}} 8 r_s \sqrt{\frac{2 r_s}{15}} \pi^{5/4} \mathrm{e}^{-(K r_s)^2/2} [3 - (K r_s)^2], \text{ and} \tag{6.85b}$$

$$p_1^1(K) = \frac{1}{\Omega} 8 r_1^2 \sqrt{\frac{r_1}{3}} \pi^{5/4} \mathrm{e}^{-(K r_s)^2/2} K. \tag{6.85c}$$

This pseudopotential can be directly incorporated into the Kohn–Sham Hamiltonian. It does not depend on the density, so if we simply want to calulate the energies and eigenfunctions of a particle moving in the pseudopotential, we just have to diagonalise the Hamiltonian which consists of the kinetic energy plus pseudopotential. A self-consistency cycle is not necessary.

The Fourier transform of the local part of the pseudopotential for a core located at $\mathbf{R}_n$ must be multiplied by $\exp(-\mathrm{i}\mathbf{K} \cdot \mathbf{R}_n)$. The nonlocal part must be multiplied by $\exp[\mathrm{i}(\mathbf{K} - \mathbf{K}') \cdot \mathbf{R}_n]$.

Check Doing this for a cubic cell with an edge length of 5 a.u. containing one hydrogen-core and an energy cut-off of $(1/2)K_{\max}^2 = 1.3$, we obtain the eigenvalues -0.03572203 (once), 0.68175686 (once), 0.80555307 (three times) and 0.83735807 (twice). If we fill all seven levels, the density should be 0.05600 on any real-space grid point. Try this first for a hydrogen at the origin, and then some arbitrary position within the cell.

6.7.6 Exchange-correlation and Hartree potentials

The exchange-correlation and Hartree potentials are density-dependent; therefore, including them makes a self-consistency cycle necessary. We shall first consider the general problem of including density-dependent potentials into the problem. First, we must have the density at our disposal. After diagonalising the Hamiltonian, we calculate the Fourier transforms $\phi^{(j)}(\mathbf{r})$ of the eigenfunctions $c_{\mathbf{K}}^{(j)}$ as in (6.63). Then we calculate the density on all real-space grid points inside the cell according to (6.66). Finally, we calculate the density in reciprocal space according to

$$n(\mathbf{K}) = \frac{1}{N^3} \sum_{\mathbf{r}} n(\mathbf{r}) \mathrm{e}^{-\mathrm{i}\mathbf{K} \cdot \mathbf{r}}. \tag{6.86}$$

For the exchange-correlation potential, we use the GTH parametrisation of the pseudopotential of Perdew and Wang [27]. This is a form of Padé approximant:

$$\varepsilon_{\mathrm{xc}} = -\frac{\sum_{i=1}^4 a_i r_s^{i-1}}{\sum_{i=1}^4 b_i r_s^i}. \tag{6.87}$$

Table 6.2. *Parameters for the GTH parametrisation of the exchange-correlation energy.*

a_1	0.4581652932831429	b_1	1.0
a_2	2.217058676663745	b_2	4.504130959426697
a_3	0.7405551735357053	b_3	1.110667363742916
a_4	0.01968227878617998	b_4	0.02359291751427506

Here, $r_s = [3/(4\pi n)]^{1/3}$ is the radius of the spherical volume per atom; the numbers a_i and b_i are given in Table 6.2. The exchange-correlation potential is given as the derivative of the energy with respect to n. It must be calculated in real space, where it is periodic (as it depends on the density, which is periodic), and then Fourier-transformed so that it can be added to the (K-space) Hamiltonian. The procedure is therefore to first fill a grid with the values of $V_{\mathrm{xc}}(\mathbf{r})$. This is then Fourier-transformed to $V_{\mathrm{xc}}(\mathbf{K})$. Then, the contribution $V_{\mathrm{xc}}(\mathbf{K}, \mathbf{K}')$, where $\mathbf{K}$ and $\mathbf{K}'$ lie inside the circle $C1$ of Figure 6.14, is found by first translating $\mathbf{K} - \mathbf{K}'$ to a point inside the unit cell UC of the reciprocal grid, and then taking for $V_{\mathrm{xc}}(\mathbf{K}, \mathbf{K}')$ the Fourier-transformed exchange-correlation potential at that reciprocal grid point.

The Hartree potential

$$V_{\mathrm{H}}(\mathbf{r}) = \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \mathrm{d}^3 r' \tag{6.88}$$

can be Fourier-transformed to give

$$V_{\mathrm{H}}(\mathbf{K}, \mathbf{K}') = V_{\mathrm{H}}(\mathbf{K} - \mathbf{K}') = \frac{4\pi}{|\mathbf{K} - \mathbf{K}'|^2} n(\mathbf{K} - \mathbf{K}'). \tag{6.89}$$

For the density, the difference $\mathbf{K} - \mathbf{K}'$ has to be translated to lie inside the unit cell UC (see Figure 6.13), just as in the case of the exchange correlation potential. For the denominator, we simply take the norm of the smallest periodic image of the difference $\mathbf{K} - \mathbf{K}'$.

Check If we incorporate both the exchange-correlation and the Hartree potentials, we have a complete self-consistent pseudopotential Kohn–Sham program. For the calculation with one hydrogen atom in a cubic cell of size 5 and a cut-off of 1.3 atomic units, we obtain the energy spectrum:

−0.468131; 0.249348; 0.373144; 0.373144;

0.373144; 0.404949; 0.404949.

6.7.7 Evaluating the energy

If you have obtained the correct spectrum, the density should necessarily be correct too. One major task remains, however: evaluating the total energy. The energy can be evaluated either by adding all the Kohn–Sham eigenvalues and subtracting the appropriate corrections as in Eq. (5.3), or by using the Kohn–Sham eigenfunctions to evaluate all the contributions to the energy as in (5.17) one by one. We take the second approach. First of all, the *kinetic energy* is given by

$$E_{\text{kin}} = \sum_j f(E_j) \sum_{\mathbf{K}} |c^{(j)}|^2 K^2/2. \tag{6.90}$$

The exchange-correlation energy is evaluated in real space:

$$E_{\text{xc}} = \sum_{\mathbf{r}} \varepsilon_{\text{xc}}(\mathbf{r}) n(\mathbf{r}), \tag{6.91}$$

where the sum is over the real-space lattice points, or in reciprocal space, where it reads:

$$E_{\text{xc}} = \Omega \sum_{\mathbf{K}} \varepsilon(\mathbf{K}) n^*(\mathbf{K}). \tag{6.92}$$

Following Marx and Hutter [25], we combine the electrostatic contributions from the electrons and the ion cores. Remember that the core part of the pseudopotential,

$$V_{\text{core}} = -\frac{Z_n}{|\mathbf{r} - \mathbf{R}_n|} \text{erf}\left(\frac{|\mathbf{r} - \mathbf{R}_n|}{\sqrt{2}\xi}\right); \tag{6.93}$$

derives from a Gaussian charge distribution:

$$n_{\text{core}}(\mathbf{r}) = -\frac{Z_n}{(\sqrt{2}\xi_n)^3} \pi^{-3/2} \exp\left[-\frac{1}{2}\left(\frac{\mathbf{r} - \mathbf{R}_n}{\xi}\right)^2\right]. \tag{6.94}$$

The Fourier transform of the core density is

$$n_{\text{core}}(\mathbf{K}) = -\frac{Z_n}{\Omega} \exp\left[-\frac{1}{2}(\xi K)^2\right] \mathrm{e}^{-\mathrm{i}\mathbf{K}\cdot\mathbf{R}_n}. \tag{6.95}$$

For the total charge density we have

$$n_{\text{tot}}(\mathbf{K}) = n_{\text{el}}(\mathbf{K}) + n_{\text{core}}(\mathbf{K}). \tag{6.96}$$

The electrostatic energy resulting from the total charge density is

$$E_{\text{ES}} = \frac{1}{2} \int \frac{n_{\text{el}}(\mathbf{r}) n_{\text{el}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, \mathrm{d}^3 r \, \mathrm{d}^3 r' + \int \frac{n_{\text{core}}(\mathbf{r}) n_{\text{el}}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} \, \mathrm{d}^3 r \, \mathrm{d}^3 r' + \frac{1}{2} \sum_{n \neq n'} \frac{Z_n Z_n'}{|\mathbf{R}_n - \mathbf{R}_n'|}, \tag{6.97}$$

where the first term is the Hartree electrostatic energy due to the electrons, the second term is the interaction between the electrons and the core, and the last term is the core–core interaction.

This last term causes problems as we must sum it over all periodic images of the unit cell for which we are performing the calculations (for a discussion concerning convergence of this type of expression, see Section 8.7.1). These problems can be avoided by replacing the last term by the electrostatic interaction of the core charges. This is done by adding and subtracting a term

$$E_{\mathrm{cc}} = \frac{1}{2}\int \frac{n_{\mathrm{core}}(\mathbf{r})n_{\mathrm{core}}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, \mathrm{d}^3r\, \mathrm{d}^3r' \tag{6.98}$$

to the expression for the total energy, Eq. (6.97). The added term, together with the first two terms in that equation, yields a contribution

$$\frac{1}{2}\int \frac{n_{\mathrm{tot}}(\mathbf{r})n_{\mathrm{tot}}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, \mathrm{d}^3r\, \mathrm{d}^3r' \tag{6.99}$$

The remaining terms can be written as a convergent sum [25], leading to

$$\begin{aligned} E_{\mathrm{ES}} = {} & \frac{1}{2}\int \frac{n_{\mathrm{tot}}(\mathbf{r})n_{\mathrm{tot}}(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, \mathrm{d}^3r\mathrm{d}^3r' \\ & + \frac{1}{2}\sum_{n,n'} \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'}|}\mathrm{erfc}\left[\frac{|\mathbf{R}_n - \mathbf{R}_{n'}|}{\sqrt{2(\xi_n^2+\xi_{n'}^2)}}\right] - \sum_n \frac{Z_n^2}{2\sqrt{\pi}\xi_n}. \end{aligned} \tag{6.100}$$

The second term on the right hand side is due to the overlap of the core distributions, and the third term corrects for the self-energy (that is, the energy of a core with itself). Both of these are contained in the first term.

For periodic boundaries, we can reformulate this expression in Fourier space, where it reads:

$$E_{\mathrm{ES}} = 2\pi\Omega \sum_{\mathbf{K}\neq\mathbf{0}} \frac{|n_{\mathrm{tot}}(\mathbf{K})|^2}{K^2} + E_{\mathrm{ovrl}} - E_{\mathrm{self}}, \tag{6.101}$$

where $n_{\mathrm{tot}}(\mathbf{K})$ is given above (Eq. (6.95)) and where

$$E_{\mathrm{ovrl}} = \sum_{\mathbf{L}} \sum_{n,n'}{}' \frac{Z_n Z_{n'}}{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|}\mathrm{erfc}\left[\frac{|\mathbf{R}_n - \mathbf{R}_{n'} - \mathbf{L}|}{\sqrt{2(\xi_n^2+\xi_{n'}^2)}}\right], \tag{6.102}$$

where $\mathbf{L}$ is an integer linear combination of the sides of the unit cell, the second sum is restricted to $n < n'$ for $\mathbf{L} = \mathbf{0}$, and

$$E_{\mathrm{self}} = \sum_n \frac{Z_n^2}{2\sqrt{\pi}\xi_n}. \tag{6.103}$$

Table 6.3. *Contributions to electronic energy of hydrogen atom.*

Contribution	*Value*
Kinetic	0.159 230 87
Short range part of pseudopotential	−0.021 083 18
Local pseudopotential	−0.244 116 10
Exchange correlation	−0.210 599 25
Hartree energy	0.024 351 53
Nonlocal pseudopotential	0.000 000 00
Local core energy	1.12582002
Self-energy	0.92508958
Electrostatic overlap	0.000 000 00
Total energy	−0.557 835 94

Finally, we must include the energy contributions due to the pseudopotential. These do not depend on the charge distribution and they contain the local and the nonlocal terms. The local contribution is easily evaluated using

$$\begin{aligned} E_{\text{local}} &= \int \sum_n V_{\text{local},n}(\mathbf{r} - \mathbf{R}_n) n(\mathbf{r})\, \mathrm{d}^3 r \\ &= \Omega \sum_n \sum_{\mathbf{K}} V_{\text{local},n}(\mathbf{K}) \mathrm{e}^{-\mathrm{i}\mathbf{K}\cdot\mathbf{R}_n} n^*(\mathbf{K}). \end{aligned} \tag{6.104}$$

where n runs over the atoms in the cell. The nonlocal energy reads:

$$E_{\text{nonlocal}} = \sum_j f_j \sum_n \sum_{l,m\varepsilon n} (F^n_{jlm})^* h^n_{lm} F^n_{jlm} \tag{6.105}$$

where $l, m\varepsilon n$ denotes the orbital with quantum numbers l, m belonging to atom number n, and

$$F^n_{jlm} = \sum_{\mathbf{K}} \mathrm{e}^{-\mathrm{i}\mathbf{K}\cdot\mathbf{R}_n} c^*_j(\mathbf{K}) Y_{lm}(\hat{\mathbf{K}}) p^l_m(K). \tag{6.106}$$

Now that you have everything in place, you can calculate the electronic energy of the hydrogen atom. It is built up from the contributions shown in Table 6.3.

6.8 Extracting information from band structures

Apart from ground state energies, from which cohesion energies and lattice spacings can be determined, and those energy levels that can be measured directly using spectroscopy experiments, it is useful to determine the density of states, $n(E)$, which can also be determined experimentally. This is defined as the number of levels between E and $E + \mathrm{d}E$, divided by $\mathrm{d}E$. Another quantity of interest is the charge density, which is needed for calculating the Hartree and exchange and correlation potentials in the DFT self-consistency loop. The charge density is given by

$$n(\mathbf{r}) = \sum_{\mathbf{k},n}{}' |\psi_{\mathbf{k},n}(\mathbf{r})|^2 = \int_{-\infty}^{E_\mathrm{F}} \mathrm{d}E \; n(\mathbf{r}, E), \tag{6.107}$$

where the sum in the second expression is over the occupied levels, i.e. those with energy below the Fermi energy E_F.

The charge density can also be found from an integration over the energy of the *local density of states*, which is defined as the charge density resulting exclusively from states at energy E. An elegant way of finding this quantity using Green's functions is described in Problem 6.3. Such an approach is necessary when the total charge of the system is not known, as is the case for a small system coupled to a large reservoir which determines the chemical potential: for a metal, this is (to very good approximation) the Fermi energy of the large system. This Fermi energy is defined with respect to the vacuum energy as the *work function*: the energy needed to remove an electron from the large system. There exist Green's function methods in which the small system (e.g. an atom or a molecule) is coupled to the surface Green's function of a metal. The electronic structure can then still be determined in a self-consistency loop, in which the charge density is determined from the Green's function of the combined system plus reservoir rather than from the eigenstates of the Hamiltonian [28].

To find physical properties or quantities, we often must perform an integration over the Brillouin zone, as the vectors (together with the band labels) in this zone are quantum numbers of the stationary states. Taking the crystal symmetry into account, these integrations only need to be carried out in the 'irreducible wedge' of the Brillouin zone: this can be used to fill the whole Brillouin zone by crystal symmetry transformations. For example, in a two-dimensional lattice having the symmetry of the square, the Brillouin zone is also a square, but to integrate quantity over the Brillouin zone, an integration over a wedge of area 1/8 of the whole square needs to be carried out. For the Brillouin zone of the fcc lattice in Figure 6.9, this irreducible wedge is the volume bounded by the labelled points.

There exist many different methods for performing Brillouin zone integration [21]. The most popular methods are those using *special points* [29, 30] and tetrahedron methods. In the latter, (part of) the Brillouin zone is divided up into tetrahedra, in each of which either a linear or a quadratic approximation of the function to be integrated is made. For calculating the density of states, the quadratic works very well since it is capable of reproducing all known Van Hove singularities [31–34].

6.9 Some additional remarks

In this chapter we have described how the nonrelativistic Schrödinger equation can be solved efficiently in a solid. The core electrons near heavy nuclei move at speeds where relativistic effects become significant, although they are still small, so that relativistic corrections must be included. This can be done in a perturbative way, and the resulting equation for the radial part $\mathcal{R}_{nl}$ of the wave function reads:

$$\left[-\frac{1}{2M}\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}^2}{\mathrm{d}r^2}\right)+\frac{l(l+1)}{2Mr^2}+V(r)-\frac{V'(r)}{4M^2c^2}\frac{\mathrm{d}}{\mathrm{d}r}\right]\mathcal{R}_{nl}(r)=E\mathcal{R}_{nl}(r). \tag{6.108}$$

$V'(r)$ is the derivative of the potential V, and M is given in terms of the electron rest mass m, the energy E and the potential as:

$$M(r)=m+\frac{1}{2c^2}[E-V(r)]. \tag{6.109}$$

This equation is derived from the Dirac equation; see for example Ref. [13].

Solving the Schrödinger equation is only one step in a DFT self-consistency equation. Having found the density as described in the previous section, we must calculate the Hartree potential by solving Poisson's equation:

$$\nabla^2 V_{\mathrm{H}}(r)=-4\pi n(\mathbf{r}). \tag{6.110}$$

Solving this equation in a pseudopotential method with a plane wave basis is not so difficult, as the Laplace operator ∇^2 has the diagonal form $k_r^2\delta_{rs}$ in reciprocal space. For muffin tins, most of the codes use a method developed by Weinert [35]. In this method we obtain an expansion of the potential in spherical harmonics. Note that in the APW method considered above, we use the spherical average of the full potential. We shall only briefly discuss the two main ideas upon which Weinert's method is based.

First of all, inside the muffin tins, the charge density and potential are expanded in spherical harmonics. The radial part of the Hartree potential can then be found by integration of a radial differential equation, as was done for the $l = 0$ case in the local density program for helium – see Section 5.5. The problem then remains of finding the solution outside the muffin tins, which is determined by the charge density in and outside the muffin tin. It seems a good idea to solve this problem in reciprocal space because of the Laplace operator being diagonal there. However, a huge number of plane waves would be necessary for obtaining an accurate solution, as the charge inside the muffin tin contains rapid oscillations (it is constructed from the wave functions which, as we have seen, vary rapidly close to the nucleus). The second ingredient of Weinert's method is the replacement of the charge density inside the spheres by a weaker one, just as in the replacement of the full potential by a pseudopotential. The new, weak charge density is called *pseudo-charge* density. That this replacement is possible can be seen by realising that the effect of a muffin tin charge distribution can be formulated in terms of the multipole moments of the charge density, and many different charge densities give the same multipole moments. Note that this justification is also analogous to the pseudopotential method, where the fact that many different potentials yield the same phase shifts justifies the replacement of the full potential by a pseudopotential. For details we refer to Weinert's paper.

6.10 Other band methods

There are numerous band structure methods [1, 10, 36]), and we have considered only two illustrative examples in this chapter. Another important approach is the Korringa–Kohn–Rostocker (KKR) method, [37–39] based on a scattering approach with a muffin tin form of potential. It leads to a matrix whose size is equal to the number of different states used in the muffin tins.

Linearising the KKR method, one obtains the linear muffin tin orbital (LMTO) method with localised, energy-independent wave functions which are centred at each atom – see Refs. [40] and [41].

Exercises

6.1 In this exercise we want to establish the relation between the energy derivative and the charge of a core wave function, Eq. (6.59). Our derivation will not rely on a spherical shape for the core region. We use the normalisation convention that the value of the wave function at the boundary of the core region is equal to some fixed number, so that we have

$$\frac{\partial \psi(\mathbf{r}_s)}{\partial E} = \dot{\psi}(\mathbf{r}_s) = 0$$

where $\mathbf{r}_s$ lies at the core boundary.

(a) Starting from the Schrödinger equation, derive an equation satisfied by $\dot{\psi}$. Note that we use the full potential, which does not depend on energy.

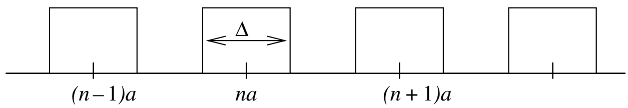
(b) Green's theorem applied to the core region for two arbitrary functions, ψ_1 and ψ_2, reads

$$\int_{\text{core}} \mathrm{d}^3 r [\psi_1(\mathbf{r}) \nabla^2 \psi_2(\mathbf{r}) - \psi_2(\mathbf{r}) \nabla^2 \psi_1(\mathbf{r})]$$
$$= \int_{\text{shell}} \mathrm{d}^2 a [\psi_1(\mathbf{a}) \hat{\mathbf{n}} \cdot \nabla \psi_2(\mathbf{a}) - \psi_2(\mathbf{a}) \hat{\mathbf{n}} \cdot \nabla \psi_1(\mathbf{a})]$$

where the integral on the right hand side is a surface integral over the boundary of the core region and $\hat{\mathbf{n}}$ is a normal vector pointing out of the core boundary. Apply this theorem to ψ and its energy derivative and use the normalisation convention to show that

$$\int_{\text{core}} \mathrm{d}^3 r \psi^2(\mathbf{r}) = -\frac{1}{2} \int_{\text{shell}} \mathrm{d}^2 a \psi(\mathbf{a}) \hat{\mathbf{n}} \cdot \nabla \dot{\psi}(\mathbf{a}).$$

6.2 [C] Consider the following periodic potential in one dimension.



The height of the barriers is V_0. The solution of the Schrödinger equation in between two barriers at $(n-1)a$ and na can be written as

$$\psi(x) = A_n \mathrm{e}^{\mathrm{i}q(x-na)} + B_n \mathrm{e}^{-\mathrm{i}q(x-na)}$$

with $q = \sqrt{2E}$. Assume that the energy we are interested in is higher than the barrier height V_0. On the nth barrier, the solution is written as

$$\psi(x) = C_n \mathrm{e}^{\mathrm{i}\kappa(x-na)} + D_n \mathrm{e}^{-\mathrm{i}\kappa(x-na)}$$

with $\kappa = \sqrt{2(E - V_0)}$.

The values of A_n and B_n in neighbouring interstitial regions are connected through the so-called 'transfer matrix':

$$\begin{pmatrix} A_{n+1} \\ B_{n+1} \end{pmatrix} = T(E) \begin{pmatrix} A_n \\ B_n \end{pmatrix}.$$

T is a 2×2 matrix which depends on energy.

(a) Show that the transfer matrix is given by

$$T = \frac{q}{4\kappa} \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix},$$

with

$$T_{11} = \mathrm{e}^{\mathrm{i}q(a-\Delta)} \left[\mathrm{e}^{\mathrm{i}\kappa\Delta} \left(1 + \frac{\kappa}{q}\right)^2 - \mathrm{e}^{-\mathrm{i}\kappa\Delta} \left(1 - \frac{\kappa}{q}\right)^2 \right],$$

$$T_{12} = -2\mathrm{i}\mathrm{e}^{\mathrm{i}qa} \left(1 - \frac{\kappa^2}{q^2}\right) \sin(\kappa\Delta);$$

and

$$T_{22} = T_{11}^*,$$
$$T_{21} = T_{12}^*.$$

Show that the product of the two eigenvalues of this matrix is equal to 1. Hence these eigenvalues can either be written as $\mathrm{e}^{\pm ik}$ (or as $\mathrm{e}^{\pm\alpha}$, real α). From Bloch's theorem we know that the solutions can be labelled by a wave vector $\tilde{q}$ which is not necessarily equal to q, and that these solutions can be written as a periodic function times $e^{i\tilde{q}x}$. In our case this implies that

$$\begin{pmatrix} A_{n+1} \\ B_{n+1} \end{pmatrix} = e^{i\tilde{q}a} \begin{pmatrix} A_n \\ B_n \end{pmatrix}$$

and therefore

$$T \begin{pmatrix} A_n \\ B_n \end{pmatrix} = e^{i\tilde{q}a} \begin{pmatrix} A_n \\ B_n \end{pmatrix}.$$

This equation defines the band spectrum of the system. It is now easier to find the vector $\tilde{q}$ as a function of energy than vice versa: the above-mentioned eigenvalues (which depend on energy via q and κ) must be equal to $e^{i\tilde{q}a}$.

(b) [C] Write a simple computer program to determine the spectrum. In an APW approach, the wave function outside the barriers is written as $e^{iq_m x}$, where $q_m = k + 2\pi m/a$ and $-\pi/a < k < \pi/a$ (m is integer). It is now convenient to confine ourselves to the unit cell $[-a/2, a/2]$ and to use Bloch boundary conditions on that cell. For a Bloch state χ_m:

$$\chi_m(-a/2) = \chi(a/2)e^{-ika}.$$

For each q_r, the value of the wave function outside and inside the barrier can be matched at the boundaries of the barrier. Show that C_m and D_m are given by

$$C_m = \frac{\sin[(\kappa + q_m)\Delta/2]}{\sin(\kappa\Delta)},$$

$$D_m = \frac{\sin[(\kappa - q_m)\Delta/2]}{\sin(\kappa\Delta)}.$$

In the APW method, the coefficients b_m of the expansion

$$\psi(x) = \sum_m b_m \chi_m(x)$$

are found by solving the generalised eigenvalue problem

$$\mathbf{Hb} = E\mathbf{Sb}$$

in which the matrix S is given by

$$\begin{aligned} S_{ml} &= \int_{-a/2}^{-\Delta/2} e^{-iq_m x} e^{iq_l x} dx + \int_{\Delta/2}^{a/2} e^{-iq_m x} e^{iq_l x} dx \\ &\quad + \int_{-\Delta/2}^{\Delta/2} [C_m^* e^{-i\kappa_m x} + D_m^* e^{i\kappa_m x}][C_l e^{i\kappa_n x} + D_l e^{-i\kappa_l x}] dx \\ &= S_{ml}^{\text{int}} + S_{ml}^{\text{ext}} \end{aligned}$$

where we have split the expression for S into an integration over the interior of the barriers and the part outside.

(c) Show that

$$S_{ml}^{\text{ext}} = \begin{cases} a - \Delta & \text{if } m = l \\ \dfrac{-2}{q_m - q_l} \sin \dfrac{(q_m - q_l)\Delta}{2} & \text{otherwise} \end{cases};$$

and

$$S_{ml}^{\text{int}} = C_m C_n + D_m D_n + (C_m D_n + C_n D_m) \frac{\kappa \Delta}{\kappa}.$$

(d) Show that the Hamiltonian matrix is given as

$$H_{ml} = H_{mn}^{\text{int}} + \frac{1}{2} q_m q_n S_{mn}^{\text{ext}} + \partial H_{mn}$$

where

$$H_{ml}^{\text{int}} = -\kappa \sin(\kappa \Delta)(C_m D_n + C_n D_m) + \kappa^2 \Delta (C_m C_n + D_m D_n),$$

and where $\partial \mathbf{H}$ is the matrix due to the jumps in derivatives across the barrier boundaries. Show that $\partial \mathbf{H}$ is given by

$$\partial H_{mn} = -q_n \sin[(q_n - q_m)\Delta/2] \\ - C_m \kappa \sin[(\kappa - q_n)\Delta/2] - D_m \kappa \sin[(\kappa + q_n)\Delta/2].$$

(e) Write a program in which the matrices H_{ml} and S_{ml} are filled and find the zeroes of the determinant $|\mathbf{H} - E\mathbf{S}|$ for various k. Compare the results with the numerically exact ones, resulting from the previous program.

6.3 In this problem we consider the determination of the local charge density using the Green's function. The Green's function for a Hamiltonian's H is defined as

$$(H - E)G(\mathbf{r}, \mathbf{r}'; E) = \delta(\mathbf{r}, \mathbf{r}').$$

(a) Show that G can be written as

$$G(\mathbf{r}, \mathbf{r}'; z) = \sum_{n=1}^{\infty} \psi_n(\mathbf{r}) \frac{1}{z - E_n} \psi_n(\mathbf{r}').$$

(b) Show that the electron density (charge density) can be found as

$$n(r) = \frac{1}{2\pi i} \int_{\Gamma} G(\mathbf{r}, \mathbf{r}; z) \mathrm{d}z,$$

where Γ is a closed contour in the complex plane which contains all the *occupied* energy levels (these of course all lie on the real axis).

6.4 [C] As plane waves form an orthogonal basis, it is possible to use the Lowdin perturbation method discussed in Section 3.4. Write an extension to your pseudopotential program to incorporate large lattice vectors into the Hamiltonian in a perturbative manner. Compare the results with those of the direct diagonalisation.

6.5 In this problem we derive the normalisation condition (6.50) from the normalisation (6.47) of the radial solution inside the muffin tin.

(a) Show that the normalisation condition (6.47) can be rewritten as

$$\langle \mathcal{R}_l | H - E | \dot{\mathcal{R}}_l \rangle = 1.$$

(b) Use this result, together with the fact that $\mathcal{R}_l$ is an eigenfunction of H inside the muffin tin with eigenvalue E, and partial differentiation, to derive Eq. (6.50).

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